

IDENTIFICATION OF AGE-DEPENDENT EFFECTS OF FUNDUS IMAGES ON INTRAOCULAR PRESSURE FOR GLAUCOMA PATIENTS: A TENSOR REGRESSION APPROACH

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Intraocular pressure (IOP) is a critical biomarker for the progression of glaucoma, which is often associated with elevated fluid pressure inside the eye. Sustained high IOP can impair the optic nerve over time, making early detection and continuous monitoring of IOP essential for glaucoma diagnosis, treatment, and disease management. A fundus image is a photograph of the back of the eye that allows doctors to see and evaluate the optic nerve head, retina, and blood vessels. Using such images to predict the IOP is a very relevant medical research topic in recent years. However, existing methods cannot capture the complicated age-dependent effects of image predictors. We thus propose a new varying coefficient model for tensor image data in this paper. We first employ a tensor partitioning strategy to reduce dimensionality, followed by a tensor decomposition technique for the high-dimensional tensor covariates. After extracting key features from the tensor covariates, we feed these low-dimensional representations into a varying coefficient model, alongside other one-way covariates. Additionally, we apply a non-concave penalty estimation to simultaneously identify the model structure and select significant predictors. A subsequent smoothing step enhances the model's accuracy. Extensive simulations are conducted to evaluate the performance of our approach. A real fundus image dataset is analyzed using our model to improve glaucoma management.

1. Introduction. Glaucoma is a chronic eye condition that can lead to optic nerve damage and vision loss and a degenerative optic neuropathy characterized by the loss of retinal ganglion cells, often leading to irreversible blindness (Jonas et al. (2017)). It affects over 64.3 million individuals worldwide, with projections suggesting an increase to 111.8 million by 2040 (Tham et al. (2014)). This heterogeneous group of disorders is the second leading cause of blindness globally, potentially impacting up to 91 million people. Glaucoma presents a wide range of risk factors, including optic nerve head (ONH) damage, visual field (VF) loss, and elevated intraocular pressure (IOP). Since glaucoma can be asymptomatic in its early and intermediate stages, diagnosis often occurs only after irreversible optic nerve damage and vision loss have already happened. Typically, the progression of glaucoma is examined by experienced ophthalmologists through retinal assessments, a process that is both tedious and time-consuming. Therefore, predicting glaucoma accurately is a critical objective for ophthalmology researchers. Recent efforts have focused on leveraging eye images to enhance predictive accuracy, which could have a significant public health impact.

Dilated fundus photography offers a convenient and inexpensive method for recording ONH structure, with glaucomatous optic neuropathy assessment remaining a gold standard for glaucoma indication. See Figure 1(a)–(b) for examples of such images from the glaucoma

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real-world appraisal progression ensemble (GRAPE) dataset (Huang et al. (2023)) which we will analyze in this paper. The longitudinal image dataset was collected at the Eye Center of the Second Affiliated Hospital of Zhejiang University. It comprises 1,115 fundus images from 263 eyes, along with associated clinical information. In particular, increased IOP is the most important biomarker for glaucoma and medical researchers are especially interested in an accurate prediction of the IOP outcome based on fundus images. Establishing such predictive models is essential for enabling timely medical intervention, which in turn can slow or prevent the progression of glaucoma in at-risk individuals.

In practice, manual assessment of the optic disc in fundus photographs requires extensive clinical training, is highly subjective with limited specialist agreement, and is time-consuming. Recent advances in AI and deep learning, along with more available data, offer promise for objective glaucoma diagnosis. However, these methods typically need large, clinically annotated datasets and are often difficult to interpret. In this work, we propose a tensor modeling technique to address these challenges.

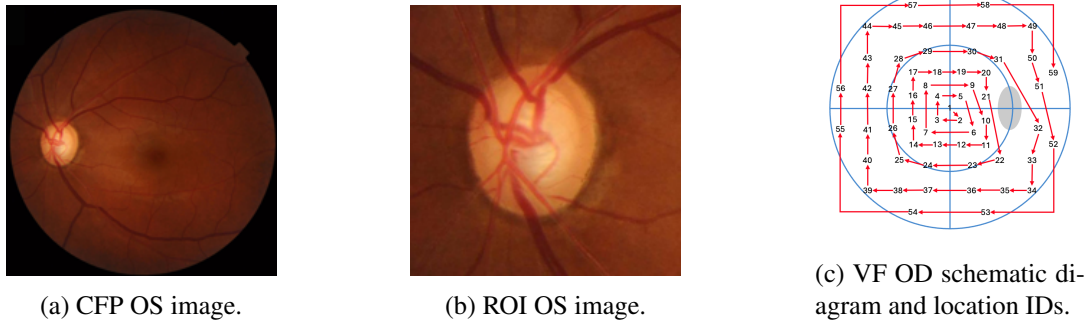


Fig 1: Representative data from the GRAPE dataset: (a) a color fundus photograph (CFP), (b) its optic-disc-centered region-of-interest (ROI) image, and (c) the visual-field (VF) schematic diagram and location IDs.

Tensors, as multidimensional generalizations of matrices, have become increasingly prominent due to their ability to represent and analyze complex, high-dimensional data structures. Extending from low-dimensional vectors and matrices, tensor objects can encapsulate data across multiple dimensions, offering a more comprehensive representation of multiway data. This flexibility makes tensors particularly valuable in many fields, such as neuroimaging (Zhou, Li and Zhu (2013); Li et al. (2018)), genomics (Hore et al. (2016)), social network analysis (Dunlavy, Kolda and Acar (2011)), hyperspectral image (Liu et al. (2019)), and recommendation system (Zhang et al. (2021)). The inherent high-dimension in tensors (Hillar and Lim (2013)) pose difficulties and challenges in real data analysis. Recent studies in statistical tensors mainly focused on tensor decomposition (Kolda and Bader (2009); Zhang and Xia (2018); Zhang and Han (2019)), tensor completion (Gandy, Recht and Yamada (2011); Yuan and Zhang (2016); Zhang (2019)), tensor regression (Zhou, Li and Zhu (2013); Zhang and Li (2017); Li et al. (2018); Zhang et al. (2020)) and tensor clustering (Han et al. (2022); Luo and Zhang (2022)).

We will consider eye images as input covariates to predict IOP using tensor regression methods. Previous authors extend traditional linear regression to tensor data, including scalar-on-tensor (Zhou, Li and Zhu (2013)), tensor-on-tensor (Lock (2018); Liu, Liu and Zhu (2020); Luo and Zhang (2022)), and tensor-on-vector (Li and Zhang (2017); Sun and Li

(2017)), among others. In these problems, the number of unknown parameters can be enormous, often far exceeding the sample size, thus imposing a significant computational burden on the estimation. To address this challenge, dimension reduction tools are necessary for an efficient estimation of tensor coefficients, including the CP decomposition-based regression (Zhou, Li and Zhu (2013); Sun and Li (2017); Bi, Qu and Shen (2018); Hao, Zhang and Cheng (2020)), the Tucker decomposition-based regression (Li et al. (2018); Guhaniyogi, Qamar and Dunson (2017); Zhang et al. (2020); Han, Willett and Zhang (2022); Luo and Zhang (2023)), and the tensor train (TT) decomposition-based regression (Liu, Liu and Zhu (2020); Liu, Lee and Zhang (2024)), among others. In addition to aforementioned low rank representation methods, one may further impose a sparsity structure and use regularization in the estimation (Zhou, Li and Zhu (2013); Zhang et al. (2020); Liu, Liu and Zhu (2020)).

Most earlier works only examined a parametric linear relationship between tensors and other variables. To capture non-linear association, varying coefficient model (VCM) offers a more flexible framework (Hastie and Tibshirani (1993); Cai, Fan and Li (2000); Fan and Zhang (1999)). There exist a large body of literature for the development of estimation methods for VCM, such as the local linear method (Zhang, Lee and Song (2002); Xia, Zhang and Tong (2004); Fan (2018); Lu (2008)), the profile least squares method (Fan and Huang (2005)), the average derivatives method (Newey and Stoker (1993)), and the smoothing spline method (Ahmad, Leelahanon and Li (2005)). VCM is especially useful for addressing dynamic and complex data problems. Cheng et al. (2014) and Cheng, Honda and Li (2016) developed a comprehensive framework of VCM in longitudinal/cluster data. Mu, Wang and Wang (2018) applied a bivariate spline on triangulation to estimate the functional coefficient of space domain data. Yu et al. (2022) used tensor-product splines to evaluate the spatio-temporal non-stationarity. Zhu, Cai and Ma (2022) studied a kernel based varying coefficient network model.

From our review, very few papers introduced the nonparametric functional effect into tensor data analysis. Chen et al. (2024) proposed a semi-parametric tensor decomposition by assuming a constant core tensor and functional factor matrices generated from tucker decomposition. Han, Shi and Zhang (2023) developed an iterative algorithm to estimate functional factor matrix along one functional axis. Zhou, Wong and He (2024) investigated a non-parametric tensor regression model and used the spline approximation by the scale-adjusted block relaxation algorithm under CP decomposition. None of these works considered varying coefficients for tensor inputs. We propose a new VCM for tensor regression and study its estimation methods in this work. In particular, we may examine the complicated age-image interactions in the glaucoma management study and demonstrate the superior prediction performance of the proposed VCM.

The varying-coefficient tensor regression model formulated in this paper can be viewed as a combination of tensor partition, tensor decomposition and varying coefficient partial linear model. Under our framework, we initially partition the whole tensor data into distinct, non-overlapping parts based on prior knowledge or distribution principles. The CP decomposition technique is applied to each partitioned tensor to extract low-rank factor representations. We only need to focus on the factor matrix along the subject dimension and with some linear algebra operation, we convert the high-dimensional functional coefficient tensor to a parsimonious functional coefficient matrix. Finally, standard nonparametric estimation methods of VCM can be implemented by vectorizing the transformed matrix covariate. Through extensive simulation studies we demonstrate the advantage of our proposed model. We finally apply this new model to examine the GRAPE data set and make some new findings for ophthalmology research.

The rest of this article is organized as follows. In Section 2, we introduce the methodology of our varying coefficient tensor regression model. In Section 3, we consider variable selection and model identification using a penalization estimation. In Section 4, simulation studies

are conducted to examine the performance of the proposed model. In Section 5, we analyze the GRAPE data using the proposed model and report our findings. Theorem, technical proofs and additional experiment results are given in the Supplementary Material.

2. Methodology.

2.1. Notations and Preliminaries. We first review some basic mathematical facts on tensor (Kolda and Bader (2009)). In this paper, we use lowercase letters (e.g. x), lowercase boldface letters (e.g. $\mathbf{x}$), uppercase boldface letters (e.g. $\mathbf{X}$), and uppercase calligraphic letters (e.g. $\mathcal{X}$) to denote scalars, vectors, matrices, and order-3-or-higher tensors respectively. An order- D **tensor** $\mathcal{X} = (x_{j_1 j_2 \dots j_D}) \in \mathbb{R}^{p_1 \times p_2 \times \dots \times p_D}$, $j_d = 1, \dots, p_d$, $d = 1, \dots, D$, is a D -dimensional array with $\prod_{d=1}^D p_d$ elements. A vector or a matrix may be regarded as order-1 or order-2 tensor respectively. We denote the **inner product** of two tensors $\mathcal{X}, \mathcal{Y} \in \mathbb{R}^{p_1 \times p_2 \times \dots \times p_D}$ as

$$(1) \quad \langle \mathcal{X}, \mathcal{Y} \rangle = \sum_{j_1=1}^{p_1} \sum_{j_2=1}^{p_2} \dots \sum_{j_D=1}^{p_D} x_{j_1, j_2, \dots, j_D} y_{j_1, j_2, \dots, j_D}$$

An **outer product** for D vectors $\mathbf{x}^{(d)} \in \mathbb{R}^{p_d}$, $d = 1, 2, \dots, D$, is an order- D tensor denoted as $\mathcal{X} = \mathbf{x}^{(1)} \circ \mathbf{x}^{(2)} \circ \dots \circ \mathbf{x}^{(D)}$, and such $\mathcal{X}$ is called rank-1 tensor.

The parallel factor analysis, also known as **CP decomposition**, factorizes a tensor into a linear combination of rank-1 tensors. In particular, a tensor satisfying a rank R CP decomposition has the following structure

$$(2) \quad \mathcal{X} = \sum_{r=1}^R \lambda_r \mathbf{x}_r^{(1)} \circ \mathbf{x}_r^{(2)} \circ \dots \circ \mathbf{x}_r^{(D)}$$

where $\mathbf{x}_r^{(d)} = (x_{j_d r}^{(d)}) \in \mathbb{R}^{p_d}$, $j_d = 1, \dots, p_d$, $d = 1, \dots, D$, $r = 1, \dots, R$, are normalized rank-1 tensor, and $\lambda_1 \geq \dots \geq \lambda_R \geq 0$ are the weights.

In this paper, suppose we observe data $\{(y_{ij}, \mathcal{X}_{ij}, \mathbf{z}_i, t_i) : i = 1, \dots, n, j = 1, 2\}$ from a sample of n units, where for the i -th unit, $y_{ij} \in \mathbb{R}$ is the response for the j th eye, $\mathcal{X}_{ij} = (x_{ij, j_1 \dots j_D}) \in \mathbb{R}^{p_1 \times \dots \times p_D}$ is an order- D tensor covariate, $\mathbf{z}_i = (z_{i,1}, \dots, z_{i,p_0})^\top \in \mathbb{R}^{p_0}$ is a p_0 -vector of scalar covariates, and $t_i \in [0, 1]$ is an index variable.

2.2. Tensor partition varying-coefficient model. Our interest is to develop a regression model for the association between the scalar response y , and its corresponding tensor covariate $\mathcal{X}$ and one-way covariates $\mathbf{z}$. We shall assume that y , $\mathcal{X}$ and $\mathbf{z}$ are centered. The regression coefficients for the tensor covariate are allowed to vary with an index variable $t \in [0, 1]$. Specifically, for the i -th unit, we assume

$$(3) \quad y_{ij} = \langle \mathcal{X}_{ij}, \mathcal{A}(t_i) \rangle + \mathbf{z}_i^\top \boldsymbol{\beta} + \epsilon_{ij}.$$

In model (3), the effect of tensor covariate on response is treated as varying tensor function $\mathcal{A}(t_i) = (\alpha_{j_1, \dots, j_D}(t_i))$, where each $\alpha_{j_1, \dots, j_D}(t_i) : [0, 1] \rightarrow \mathbb{R}$ is an unknown mapping. The coefficient for one-way covariate $\mathbf{z}_i$ is a constant vector $\boldsymbol{\beta} \in \mathbb{R}^{p_0}$. **We assume that different paired units are independent and that, within paired unit i , the errors have mean zero and covariance**

$$(4) \quad \boldsymbol{\Sigma}_i = \sigma^2(t_i) \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}, \quad |\rho| < 1,$$

where $\sigma^2(t) > 0$ is an unknown smooth variance function and ρ is a common within-pair correlation. In this model, the index variable t_i can be any continuous covariate. For example,

in the glaucoma study, investigators are interested in examining how the effects of fundus images on the intraocular pressure change with age and we thus choose t to be the patient's age in this study.

A commonly observed image tensor $\mathcal{X}$ would require the estimation of enormous unknown varying-coefficients. It is computationally prohibitive to directly estimate $\mathcal{A}(t)$ due to the high dimensionality. However, not all elements of $\mathcal{X}$ contribute to the response of interest. Often, regions of interest are confined to a small portion of the entire tensor. Therefore, it is more sensible to focus on sub-tensors that capture those local features. To this end, we consider a partition model that divides the concatenated full-sample tensor $\tilde{\mathcal{X}}$ into S disjoint sub-tensor covariates $\tilde{\mathcal{X}}^{(s)}$, that is

$$(5) \quad \tilde{\mathcal{X}} = \cup_{s=1}^S \tilde{\mathcal{X}}^{(s)} \quad \text{and} \quad \tilde{\mathcal{X}}^{(s)} \cap \tilde{\mathcal{X}}^{(s')} = \emptyset \quad \forall s \neq s'.$$

Without loss of generality, we assume that the size of each sub-tensor $\tilde{\mathcal{X}}^{(s)} \in \mathbb{R}^{2n \times p'_1 \times \dots \times p'_D}$ is fixed for any s such that $S = \prod_{d=1}^D (p_d/p'_d) \equiv \prod_{d=1}^D S^{(d)}$, where $S^{(d)}$ is the number of partitions along the d -th order. We denote $\mathcal{X}_{ij}^{(s)} \in \mathbb{R}^{p'_1 \times \dots \times p'_D}$ as partition s of eye j in paired unit i , and $\tilde{\mathcal{X}}^{(s)} = \{\mathcal{X}_{ij}^{(s)} : i = 1, \dots, n, j = 1, 2\}$.

Corresponding to the partition pattern of $\tilde{\mathcal{X}}$, we can also divide $\mathcal{A}(t)$ into S disjoint sub-tensor functions $\mathcal{A}^{(s)}(t)$, that is

$$(6) \quad \mathcal{A}(t) = \cup_{s=1}^S \mathcal{A}^{(s)}(t) \quad \text{and} \quad \mathcal{A}^{(s)}(t) \cap \mathcal{A}^{(s')}(t) = \emptyset \quad \forall s \neq s'.$$

With such tensor partition technique, we can rewrite the tensor effect in (3) as $\langle \mathcal{X}_{ij}, \mathcal{A}(t_i) \rangle = \sum_{s=1}^S \langle \mathcal{X}_{ij}^{(s)}, \mathcal{A}^{(s)}(t_i) \rangle$.

Next we adopt a rank- R CP decomposition model on each partition to further reduce the sub-tensor $\tilde{\mathcal{X}}^{(s)}$ to lower dimensional feature matrices. Specifically, we have

$$(7) \quad \tilde{\mathcal{X}}^{(s)} \approx \sum_{r=1}^R \lambda_r^{(s)} \mathbf{x}_{rs}^* \circ \mathbf{x}_r^{(s1)} \circ \dots \circ \mathbf{x}_r^{(sD)},$$

where $\lambda_r^{(s)}$ is the weight of the r -th rank of the decomposition, $\mathbf{x}_r^{(sd)} = (x_{1r}^{(sd)}, \dots, x_{p'_d r}^{(sd)})^\top \in \mathbb{R}^{p'_d}$ is the common factor along the d -th order of the r -th rank, and $\mathbf{x}_{rs}^* = (x_{11rs}^*, x_{12rs}^*, \dots, x_{n1rs}^*, x_{n2rs}^*)^\top \in \mathbb{R}^{2n}$ is the factor along the eye-within-pair dimension. The elements of $\mathbf{x}_{rs}^*$ contain the primary variation in the tensor variables due to eye and paired-unit differences, while the common structure is absorbed into factors $\mathbf{x}_r^{(sd)}$, $r = 1, \dots, R$, $d = 1, \dots, D$ (Miranda et al. (2018)). For the tensor covariate of eye j in paired unit i , (7) gives

$$(8) \quad \mathcal{X}_{ij}^{(s)} \approx \sum_{r=1}^R \lambda_r^{(s)} x_{ijrs}^* \mathbf{x}_r^{(s1)} \circ \dots \circ \mathbf{x}_r^{(sD)} \equiv \sum_{r=1}^R x_{ijrs}^* \mathcal{U}_r^{(s)},$$

where $\mathcal{U}_r^{(s)} = \lambda_r^{(s)} \mathbf{x}_r^{(s1)} \circ \dots \circ \mathbf{x}_r^{(sD)} \in \mathbb{R}^{p'_1 \times \dots \times p'_D}$. We note that $\mathcal{U}_r^{(s)}$ contains only the shared information among subjects and thus can be dropped from the regression analysis.

After the above partitioning and decomposition, essentially only x_{ijrs}^* may contribute to the response variation and thus serve as an appropriate covariate. Consequently, the total number of measurements for an individual tensor is reduced from $p_1 \times \dots \times p_D$ to $R \times S$. A similar method is adopted in Tang, Bi and Qu (2020) to learn individualized tensor construction. If we denote $a_{rs}(t) \equiv \langle \mathcal{U}_r^{(s)}, \mathcal{A}^{(s)}(t) \rangle$ as the corresponding coefficient function of x_{ijrs}^* , $i = 1, \dots, n$, $j = 1, 2$, and $\mathbf{A}(t) = (a_{rs}(t))$, model (3) now becomes

$$y_{ij} = \langle \mathcal{X}_{ij}, \mathcal{A}(t_i) \rangle + \mathbf{z}_i^\top \boldsymbol{\beta} + \epsilon_{ij}$$

$$\begin{aligned}
&= \sum_{s=1}^S \langle \mathcal{X}_{ij}^{(s)}, \mathcal{A}^{(s)}(t_i) \rangle + \mathbf{z}_i^\top \boldsymbol{\beta} + \epsilon_{ij} \\
&\approx \sum_{s=1}^S \left\langle \sum_{r=1}^R x_{ijrs}^* \mathcal{U}_r^{(s)}, \mathcal{A}^{(s)}(t_i) \right\rangle + \mathbf{z}_i^\top \boldsymbol{\beta} + \epsilon_{ij} \\
&= \sum_{s=1}^S \sum_{r=1}^R x_{ijrs}^* \langle \mathcal{U}_r^{(s)}, \mathcal{A}^{(s)}(t_i) \rangle + \mathbf{z}_i^\top \boldsymbol{\beta} + \epsilon_{ij} \\
(9) \quad &= \langle \mathbf{X}_{ij}^*, \mathbf{A}(t_i) \rangle + \mathbf{z}_i^\top \boldsymbol{\beta} + \epsilon_{ij},
\end{aligned}$$

where $\mathbf{X}_{ij}^* = (x_{ijrs}^*) \in \mathbb{R}^{R \times S}$. Throughout this paper, we shall assume that R and S are constants. Instead of applying the decomposition on coefficient tensor $\mathcal{A}$ (Zhou, Li and Zhu (2013); Zhang and Li (2017); Li et al. (2018)), we place the low-rank assumption on the tensor covariate $\mathcal{X}$ and make no specific assumptions regarding the structure of the tensor function $\mathcal{A}(t)$. Furthermore, we do not need more assumptions on tensor covariate $\mathcal{X}$, such as Gaussian ensemble design (Zhang et al. (2020); Han, Willett and Zhang (2022)) or tensor restricted isometry property (Luo and Zhang (2023); Tong et al. (2022)). The model considered in this paper thus presents a novel introduction to the literature.

2.3. Estimating functions and parameters. For the unknown functional coefficients, we only require the usual smoothness condition which enables the local linear approximation by a first order Taylor expansion. Specifically, for t_i is in a small neighborhood of t , we have

$$(10) \quad \mathbf{A}(t_i) \approx \mathbf{A}(t) + \dot{\mathbf{A}}(t)(t_i - t),$$

where $\dot{\mathbf{A}}(t)$ is the first derivative of $\mathbf{A}(t)$. We first obtain the independence estimators of the unknown functions $\mathbf{A}(t)$ and parameters $\boldsymbol{\beta}$ by treating the two eyes as independent and minimizing the local least squares

$$\begin{aligned}
&\sum_{i=1}^n \sum_{j=1}^2 [y_{ij} - \mathbf{v}_{ij}^\top (\mathbf{a}_t + \mathbf{b}_t(t_i - t)) - \mathbf{z}_i^\top \boldsymbol{\beta}]^2 K_h(t_i - t) \\
(11) \quad &= \sum_{i=1}^n \sum_{j=1}^2 [y_{ij} - \tilde{\mathbf{v}}_{ij}^\top(t) \boldsymbol{\theta}_t]^2 K_h(t_i - t),
\end{aligned}$$

where $\mathbf{v}_{ij} = \text{vec}(\mathbf{X}_{ij}^*) \in \mathbb{R}^{RS}$, $\mathbf{a}_t = \text{vec}(\mathbf{A}(t))$, $\mathbf{b}_t = \text{vec}(\dot{\mathbf{A}}(t))$, $\tilde{\mathbf{v}}_{ij}^\top(t) = (\mathbf{z}_i^\top, \mathbf{v}_{ij}^\top, (t_i - t)\mathbf{v}_{ij}^\top)^\top \in \mathbb{R}^{2RS+p_0}$, $\boldsymbol{\theta}_t = (\boldsymbol{\beta}^\top, \mathbf{a}_t^\top, \mathbf{b}_t^\top)^\top$ and $K_h(t) = K(t/h)/h$ is a scaled kernel with bandwidth h . It is straightforward to derive the closed-form solution of (11) to be

$$(12) \quad \hat{\boldsymbol{\theta}}_t^\dagger = [\tilde{\mathbf{V}}^\top(t) \mathbf{W}(t) \tilde{\mathbf{V}}(t)]^{-1} \tilde{\mathbf{V}}^\top(t) \mathbf{W}(t) \mathbf{y},$$

where $\mathbf{y} = (y_{11}, y_{12}, \dots, y_{n1}, y_{n2})^\top \in \mathbb{R}^{2n}$, $\tilde{\mathbf{V}}(t) = (\tilde{\mathbf{v}}_{11}(t), \dots, \tilde{\mathbf{v}}_{n2}(t))^\top \in \mathbb{R}^{2n \times (2RS+p_0)}$, and $\mathbf{W}(t) = \text{diag}(K_h(t_1 - t), K_h(t_1 - t), \dots, K_h(t_n - t)) \in \mathbb{R}^{2n \times 2n}$. We thus have vectorized estimate $\text{vec}(\hat{\mathbf{A}}^\dagger(t)) = \hat{\mathbf{a}}_t^\dagger = (\mathbf{0}_{RS \times p_0}, \mathbf{I}_{RS}, \mathbf{0}_{RS \times RS}) \hat{\boldsymbol{\theta}}_t^\dagger$. In practice, the bandwidth h is chosen by the leave-one-out cross validation.

After the independence function estimate $\hat{\mathbf{A}}^\dagger(t)$ is attained, we may evaluate the corresponding global estimator of $\boldsymbol{\beta}$ by minimizing

$$(13) \quad \sum_{i=1}^n \sum_{j=1}^2 [y_{ij} - \langle \mathbf{X}_{ij}^*, \hat{\mathbf{A}}^\dagger(t_i) \rangle - \mathbf{z}_i^\top \boldsymbol{\beta}]^2.$$

The independence estimate $\hat{\beta}^\dagger$ is given by

$$(14) \quad \hat{\beta}^\dagger = [\mathbf{Z}^\top \mathbf{Z}]^{-1} \mathbf{Z} \mathbf{y}^\dagger,$$

where $\mathbf{Z} = (\mathbf{z}_1, \mathbf{z}_1, \dots, \mathbf{z}_n)^\top \in \mathbb{R}^{2n \times p_0}$, and $\mathbf{y}^\dagger = (y_{11} - \langle \mathbf{X}_{11}^*, \hat{\mathbf{A}}^\dagger(t_1) \rangle, \dots, y_{n2} - \langle \mathbf{X}_{n2}^*, \hat{\mathbf{A}}^\dagger(t_n) \rangle)^\top \in \mathbb{R}^{2n}$.

The independence estimators ignore the correlation between observations from the two eyes. We therefore propose paired estimators that use the within-subject covariance structure to improve efficiency. If $\Sigma(t)$ is known, these estimators are

$$(15) \quad \hat{\theta}_t^* = [\tilde{\mathbf{V}}^\top(t) \Sigma^{-1/2}(t) \mathbf{W}(t) \Sigma^{-1/2}(t) \tilde{\mathbf{V}}(t)]^{-1} \tilde{\mathbf{V}}^\top(t) \Sigma^{-1/2}(t) \mathbf{W}(t) \Sigma^{-1/2}(t) \mathbf{y},$$

and

$$(16) \quad \hat{\beta}^* = [\mathbf{Z}^\top \Sigma^{-1}(t) \mathbf{Z}]^{-1} \mathbf{Z} \Sigma^{-1}(t) \mathbf{y}^\dagger,$$

if the covariance $\Sigma(t) = \mathbf{I}_n \otimes \Sigma_0(t)$ is known. To this end, we may estimate the covariance parameters by

$$(17) \quad \hat{\sigma}^2(t) = \frac{\sum_{i=1}^n \sum_{j=1}^2 [y_{ij} - \langle \mathbf{X}_{ij}^*, \hat{\mathbf{A}}^\dagger(t_i) \rangle - \mathbf{z}_i^\top \hat{\beta}^\dagger]^2 K_h(t_i - t)}{2 \sum_{i=1}^n K_h(t_i - t)},$$

where h is a bandwidth for smoothing the variance function and

$$(18) \quad \hat{\rho} = \sum_{i=1}^n [y_{i1} - \langle \mathbf{X}_{i1}^*, \hat{\mathbf{A}}^\dagger(t_i) \rangle - \mathbf{z}_i^\top \hat{\beta}^\dagger] [y_{i2} - \langle \mathbf{X}_{i2}^*, \hat{\mathbf{A}}^\dagger(t_i) \rangle - \mathbf{z}_i^\top \hat{\beta}^\dagger] / \sum_{i=1}^n \hat{\sigma}^2(t_i).$$

We thus replace $\sigma^2(t)$ and ρ in (15) and (16) by (17) and (18) and obtain the feasible paired estimators $\hat{\theta}_t^*$ and $\hat{\beta}^*$.

Remark 1 Under conditions (C1)–(C4) in the Supplementary, we show in Section C.1 that the local linear estimators for the time-varying coefficient matrix $A(t)$ and its derivative are jointly asymptotically normal with standard kernel-smoothing bias and variance of order $(nh)^{-1}$; the optimal bandwidth for estimating $A(t)$ is $h \sim n^{-1/5}$ (c.f. Theorem 1 in Section C.1). To achieve $\sqrt{n}$ -consistency and asymptotic normality for the parametric component β , undersmoothing is required (i.e., choosing h to be smaller than $n^{-1/5}$ so that $nh^4 \rightarrow 0$ and $nh^3 / \log n \rightarrow \infty$, c.f. Theorem 2 in Section C.1).

3. Variable selection and structure identification.

3.1. Group penalty procedure. Even after tensor partition and decomposition, there are still RS functions to be estimated in $\mathbf{A}(t) = (a_{rs}(t))$, $r = 1, \dots, R$, $s = 1, \dots, S$, in addition to p_0 Euclidean parameters in β . The dimension may still be quite large relative to the available sample size. In practice not all these components are effective to the response and a sparsity structure is usually plausible. Furthermore, not all functional coefficients are varying and need to be estimated nonparametrically. We further adopt a regularization procedure for variable selection and model structure identification.

To implement the shrinkage approach, we consider spline basis approximation method (De Boor (2001)) for functional estimation in this section. While the kernel method can usually provide a better local approximation to the function estimation, it is often less efficient suitable for the penalized computations. In contrast, the spline method is computationally more efficient for our purpose of regularization.

For a chosen set of B-splines basis functions $\{B_l(t), l = 1, \dots, L\}$, we write $\mathbf{B}(t) = (B_1(t), \dots, B_L(t))^T$ where each $B_l(t)$ is a B-spline function, $L = k_n + k_d + 1$ is the number

of basis functions, k_n is the number of equispaced internal knots located in $[0, 1]$, and k_d is the degree of the polynomial spline. The spline approximation of the varying coefficient can be written in terms of B-splines as

$$(19) \quad a_{rs}(t) \approx \sum_{l=1}^L \gamma_{lrs} B_l(t) = \mathbf{B}^\top(t) \boldsymbol{\gamma}_{rs},$$

where $\boldsymbol{\gamma}_{rs} = (\gamma_{1rs}, \dots, \gamma_{Lrs})^T$ are some suitable coefficients.

Here we consider B-spline basis for its desirable properties such as the numerical stability (De Boor (2001)) while other basis functions can be used for a similar purpose. We need to choose k_n to control the smoothness of nonparametric function $a_{rs}(t)$'s and require $k_n \rightarrow \infty$ as $n \rightarrow \infty$. Although it is possible to allow different number of basis for different functions $a_{rs}(t)$, we specify them all equal to L for the simplicity of notation.

After substituting (19) into (9), we obtain a slightly different tensor regression form:

$$(20) \quad y_{ij} \approx \sum_{l=1}^L \sum_{r=1}^R \sum_{s=1}^S x_{ijrs}^* B_l(t_i) \gamma_{lrs} + \mathbf{z}_i^\top \boldsymbol{\beta} + \epsilon_{ij} \equiv \langle \mathcal{H}_{ij}(t_i), \mathcal{G} \rangle + \mathbf{z}_i^\top \boldsymbol{\beta} + \epsilon_{ij},$$

where $\mathcal{H}_{ij}(t) = \mathbf{X}_{ij}^* \circ \mathbf{B}(t) = (x_{ijrs}^* B_l(t)) \equiv (h_{ijlrs}(t)) \in \mathbb{R}^{L \times R \times S}$, and $\mathcal{G} = (\gamma_{lrs}) \in \mathbb{R}^{L \times R \times S}$.

Before we define the penalty term in a projected B-spline space for simultaneous variable selection and structure identification, we introduce an orthogonal decomposition of $a_{rs}(t)$ with respect to the L_2 norm by

$$(21) \quad a_{rs}(t) = (a_{rs})_c + (a_{rs})_v(t),$$

where $(a_{rs})_c = \int_0^1 a_{rs}(t) dt \approx \int_0^1 \mathbf{B}^\top(t) dt \boldsymbol{\gamma}_{rs} \in \mathbb{R}$ is the mean of $a_{rs}(t)$, and $(a_{rs})_v(t) = a_{rs}(t) - (a_{rs})_c \approx (\mathbf{B}^\top(t) - \int_0^1 \mathbf{B}^\top(t) dt) \boldsymbol{\gamma}_{rs}$ is the deviation from the mean. Then we have

$$(22) \quad \|a_{rs}\|_2^2 = |(a_{rs})_c|^2 + \|(a_{rs})_v\|_2^2,$$

where $|\cdot|$ denotes the absolute value, and $\|\cdot\|_2$ is the L_2 norm. The two terms on the right hand side of (22) can be easily evaluated by $|(a_{rs})_c|^2 \approx \boldsymbol{\gamma}_{rs}^\top \int_0^1 \mathbf{B}(t) dt \int_0^1 \mathbf{B}^\top(t) dt \boldsymbol{\gamma}_{rs}$, and $\|(a_{rs})_v\|_2^2 \approx \boldsymbol{\gamma}_{rs}^\top [\int_0^1 (\mathbf{B}(t) - \int_0^1 \mathbf{B}(t) dt) (\mathbf{B}^\top(t) - \int_0^1 \mathbf{B}^\top(t) dt) dt] \boldsymbol{\gamma}_{rs} = \boldsymbol{\gamma}_{rs}^\top (\int_0^1 \mathbf{B}(t) \mathbf{B}^\top(t) dt - \int_0^1 \mathbf{B}(t) dt \int_0^1 \mathbf{B}^\top(t) dt) \boldsymbol{\gamma}_{rs}$. It is clear that $\|(a_{rs})_v\|_2 \neq 0$ when a_{rs} is a varying function, $\|(a_{rs})_v\|_2 = 0$ when a_{rs} is a constant function, and furthermore, $|(a_{rs})_c| = \|(a_{rs})_v\|_2 = 0$ when a_{rs} is a constant zero function.

We then propose the following penalized loss function to simultaneously select important variables and identify the model structure with group regularized estimator $\hat{\mathcal{G}}^\ddagger$ and $\hat{\boldsymbol{\beta}}^\ddagger$:

$$(23) \quad \begin{aligned} & \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^2 [y_{ij} - \langle \mathcal{H}_{ij}(t_i), \mathcal{G} \rangle - \mathbf{z}_i^\top \boldsymbol{\beta}]^2 + n \sum_{k=1}^{p_0} P_{\omega_\beta}(|\beta_k|) \\ & + n \sum_{r=1}^R \sum_{s=1}^S \{P_{\omega_\gamma^c}(|(a_{rs})_c|) + P_{\omega_\gamma^v}(\|(a_{rs})_v\|_2)\}, \end{aligned}$$

where $P_\omega(\cdot)$'s are all SCAD penalty functions with a group tuning parameter ω . In general the SCAD function is defined through its first derivative as

$$(24) \quad \dot{P}_\omega(\theta) = \omega \left[I\{|\theta| \leq \omega\} + \frac{(a_0 \omega - |\theta|)_+}{(a_0 - 1)\omega} I\{|\theta| > \omega\} \right]$$

with $a_0 > 2$, where $I\{\cdot\}$ is an indicator function. In our numerical studies, we set $a_0 = 3.7$ as suggested by [Fan and Li \(2001\)](#).

To implement the above minimization, following [Fan and Li \(2001\)](#), we use an iterative local quadratic approximation algorithm to solve the penalized estimation. Using the Taylor expansion, given an estimator of θ in $(c-1)$ -th iteration, $\theta^{(c-1)}$, we can approximate the regularization term $P_\omega(\theta)$ by

$$(25) \quad P_\omega(\theta) \approx P_\omega(\theta^{(c-1)}) + \frac{1}{2} \frac{\dot{P}_\omega(\theta^{(c-1)})}{\theta^{(c-1)}} (\theta^2 - (\theta^{(c-1)})^2).$$

The iterative algorithm may be initialized at $\mathcal{G}^{(0)}$ and $\beta^{(0)}$ using the independence estimates $\mathbf{A}^\dagger(t)$ and $\beta^\dagger$ in (12) and (14), or the least squares estimates from (23) with $\omega_\beta = \omega_\gamma^c = \omega_\gamma^v = 0$.

In the c -th iteration, we first update $\beta^{(c)}$ by minimizing

$$(26) \quad \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^2 [y_{ij} - \langle \mathcal{H}_{ij}(t_i), \mathcal{G}^{(c-1)} \rangle - \mathbf{z}_i^\top \beta]^2 + n \sum_{k=1}^{p_0} P_{\omega_\beta}(|\beta_k|)$$

where $\mathcal{G}^{(c-1)}$ is the estimate of $\mathcal{G}$ in the $(c-1)$ -th iteration. The approximate solution of (26) is

$$(27) \quad \beta^{(c)} = \left(\mathbf{Z}^\top \mathbf{Z} + n \Omega_0(\beta^{(c-1)}, \omega_\beta) \right)^{-1} \mathbf{Z}^\top \mathbf{y}_\mathbf{Z}^{(c-1)},$$

where $\mathbf{y}_\mathbf{Z}^{(c-1)} = (y_{11} - \langle \mathcal{H}_{11}(t_1), \mathcal{G}^{(c-1)} \rangle, \dots, y_{n2} - \langle \mathcal{H}_{n2}(t_n), \mathcal{G}^{(c-1)} \rangle)^\top \in \mathbb{R}^{2n}$, $\mathbf{Z} = (\mathbf{z}_1, \dots, \mathbf{z}_n)^\top \in \mathbb{R}^{2n \times p_0}$, $\Omega_0(\beta^{(c-1)}, \omega_\beta) = \text{diag}\left(\frac{\dot{P}_{\omega_\beta}(|\beta_1^{(c-1)}|)}{|\beta_1^{(c-1)}|}, \dots, \frac{\dot{P}_{\omega_\beta}(|\beta_{p_0}^{(c-1)}|)}{|\beta_{p_0}^{(c-1)}|}\right) \in \mathbb{R}^{p_0 \times p_0}$ is a diagonal matrix, and $\beta^{(c-1)} = (\beta_1^{(c-1)}, \dots, \beta_{p_0}^{(c-1)}) \in \mathbb{R}^{p_0}$ is the estimate of β in the $(c-1)$ -th iteration.

Then, given the current estimate $\beta^{(c)}$, we update the estimate $\mathcal{G}^{(c)}$ by minimizing

$$(28) \quad \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^2 [y_{ij} - \langle \mathcal{H}_{ij}(t_i), \mathcal{G} \rangle - \mathbf{z}_i^\top \beta^{(c)}]^2 + n \sum_{r=1}^R \sum_{s=1}^S P_{\omega_\gamma^c}(|(a_{rs})_c|) + n \sum_{r=1}^R \sum_{s=1}^S P_{\omega_\gamma^v}(\|(a_{rs})_v\|_2).$$

Similar to (27), $\mathcal{G}^{(c)}$ can be obtained through

$$(29) \quad \text{vec}(\mathcal{G}^{(c)}) = \left(\mathbf{H}^\top \mathbf{H} + n \Omega_1(\mathcal{G}^{(c-1)}, \omega_\gamma^c) + n \Omega_2(\mathcal{G}^{(c-1)}, \omega_\gamma^v) \right)^{-1} \mathbf{H}^\top \mathbf{y}_\mathbf{H}^{(c)},$$

where $\mathbf{y}_\mathbf{H}^{(c)} = (y_{11} - \mathbf{z}_1^\top \beta^{(c)}, \dots, y_{n2} - \mathbf{z}_n^\top \beta^{(c)})^\top \in \mathbb{R}^{2n}$, $\mathbf{H} = (\text{vec}(\mathcal{H}_{11}(t_1)), \dots, \text{vec}(\mathcal{H}_{n2}(t_n)))^\top \in \mathbb{R}^{2n \times LRS}$, and $\text{vec}(\mathcal{H}_{ij}(t_i)) = (h_{ijlrs}(t_i)) = (h_{ij111}(t_i), \dots, h_{ijL11}(t_i), \dots, h_{ijLR1}(t_i), \dots, h_{ijLRS}(t_i))^\top \in \mathbb{R}^{LRS}$. As for two penalized terms, we have

$$\Omega_1(\mathcal{G}^{(c-1)}, \omega_\gamma^c) = \text{diag}\left(\frac{\dot{P}_{\omega_\gamma^c}(|(a_{11}^{(c-1)})_c|)}{|(a_{11}^{(c-1)})_c|} \mathcal{B}_1, \dots, \frac{\dot{P}_{\omega_\gamma^c}(|(a_{R1}^{(c-1)})_c|)}{|(a_{R1}^{(c-1)})_c|} \mathcal{B}_1, \dots, \frac{\dot{P}_{\omega_\gamma^c}(|(a_{RS}^{(c-1)})_c|)}{|(a_{RS}^{(c-1)})_c|} \mathcal{B}_1\right)$$

is an LRS -by- LRS block diagonal matrix, where $|(a_{rs}^{(c-1)})_c| = \sqrt{\gamma_{rs}^{(c-1)\top} \mathcal{B}_1 \gamma_{rs}^{(c-1)}}$ with $\gamma_{rs} = (\gamma_{lrs}) \in \mathbb{R}^L$, and $\mathcal{B}_1 \in \mathbb{R}^{L \times L}$ is a matrix given by

$$\mathcal{B}_1 = \int_0^1 \mathbf{B}(t) dt \int_0^1 \mathbf{B}^\top(t) dt = \begin{bmatrix} (\int_0^1 B_1(t) dt)^2 & \cdots & \int_0^1 B_1(t) dt \int_0^1 B_L(t) dt \\ \vdots & \ddots & \vdots \\ \int_0^1 B_L(t) dt \int_0^1 B_1(t) dt & \cdots & (\int_0^1 B_L(t) dt)^2 \end{bmatrix},$$

and

$$\Omega_2(\mathcal{G}^{(c-1)}, \omega_\gamma^v) = \text{diag}\left(\frac{\dot{P}_{\omega_\gamma^v}(\|(a_{11}^{(c-1)})_v\|_2)}{\|(a_{11}^{(c-1)})_v\|_2} \mathcal{B}_2, \dots, \frac{\dot{P}_{\omega_\gamma^v}(\|(a_{R1}^{(c-1)})_v\|_2)}{\|(a_{R1}^{(c-1)})_v\|_2} \mathcal{B}_2, \dots, \frac{\dot{P}_{\omega_\gamma^v}(\|(a_{RS}^{(c-1)})_v\|_2)}{\|(a_{RS}^{(c-1)})_v\|_2} \mathcal{B}_2\right)$$

is also an *LRS-by-LRS* block diagonal matrix, where $\|(a_{rs}^{(c-1)})_v\|_2 = \sqrt{\gamma_{rs}^{(c-1)\top} \mathcal{B}_2 \gamma_{rs}^{(c-1)}}$ and $\mathcal{B}_2 \in \mathbb{R}^{L \times L}$ is a matrix given by

$$\begin{aligned} \mathcal{B}_2 &= \int_0^1 \mathbf{B}(t) \mathbf{B}^\top(t) dt - \int_0^1 \mathbf{B}(t) dt \int_0^1 \mathbf{B}^\top(t) dt \\ &= \begin{bmatrix} \int_0^1 B_1^2(t) dt - (\int_0^1 B_1(t) dt)^2 & \dots & \int_0^1 B_1(t) B_L(t) dt - \int_0^1 B_1(t) dt \int_0^1 B_L(t) dt \\ \vdots & \ddots & \vdots \\ \int_0^1 B_L(t) B_1(t) dt - \int_0^1 B_L(t) dt \int_0^1 B_1(t) dt & \dots & \int_0^1 B_L^2(t) dt - (\int_0^1 B_L(t) dt)^2 \end{bmatrix}. \end{aligned}$$

The vector $\text{vec}(\mathcal{G}^{(c)})$ can be easily tensorized to produce the corresponding $\mathcal{G}^{(c)}$.

We iteratively update β and $\mathcal{G}$ until convergence, that is $\|\beta^{(c)} - \beta^{(c-1)}\|_2^2 + \|\mathcal{G}^{(c)} - \mathcal{G}^{(c-1)}\|_2^2 < 10^{-8}$ in this work. Finally, we set $\hat{\mathcal{G}}^\dagger = \mathcal{G}^{(c)}$ and $\hat{\beta}^\dagger = \beta^{(c)}$ upon convergence. We may obtain the penalized tensor function $\hat{\mathbf{A}}^\dagger(t) = (\hat{a}_{rs}^\dagger(t)) = (\mathbf{B}^\top(t) \hat{\gamma}_{rs}^\dagger)$.

Remark 2 Instead of shrinking $|(a_{rs})_c|$ and $\|(a_{rs})_v\|_2$ to zero, we could also operationally replace them by $|(\gamma_{rs})_c|$ and $\|(\gamma_{rs})_v\|_2$ where $(\gamma_{rs})_c = \frac{1}{L} \sum_{l=1}^L \gamma_{lrs}$ is the mean and $(\gamma_{rs})_v = \gamma_{rs} - (\gamma_{rs})_c \mathbf{1} \in \mathbb{R}^L$ represents the deviation function. These two pairs are quite similar in both theory and practice since we can choose the order of spline to ensure a one-to-one corresponding projection. More specifically, we can expand $|(\gamma_{rs})_c|^2 = \frac{1}{L^2} \gamma_{rs}^\top \mathbf{1} \mathbf{1}^\top \gamma_{rs}$, and $\|(\gamma_{rs})_v\|_2^2 = \gamma_{rs}^\top (\mathbf{I} - \frac{1}{L} \mathbf{1} \mathbf{1}^\top) (\mathbf{I} - \frac{1}{L} \mathbf{1} \mathbf{1}^\top)^\top \gamma_{rs} = \gamma_{rs}^\top (\mathbf{I} - \frac{1}{L} \mathbf{1} \mathbf{1}^\top) \gamma_{rs}$, where $\mathbf{1} \in \mathbb{R}^L$ is a vector of ones and $\mathbf{I} \in \mathbb{R}^{L \times L}$ is an identity matrix.

Remark 3 To identify constant coefficients in $\mathbf{A}(t)$, we can also use the term $\|\dot{a}_{rs}\|_2 = \sqrt{\gamma_{rs}^\top \dot{\mathcal{B}} \gamma_{rs}}$ to replace $\|(a_{rs})_v\|_2$, where $\dot{\mathcal{B}} = \{\int_0^1 \dot{\mathbf{B}}_l(t)^\top \dot{\mathbf{B}}_{l'}(t) dt : l, l' = 1, \dots, L\} \in \mathbb{R}^{L \times L}$. The idea is to shrink the first derivative $\|\dot{a}_{rs}\|_2$ to zero, then $a_{rs}(t)$ is a constant function. This penalty term will produce a similar result as $\|(a_{rs})_v\|_2$ (Guo and Li (2022)).

Remark 4 In numerical studies, we have also used Lasso (Tibshirani (1996)) and MCP (Zhang (2010)) penalty functions in place of $P_\omega(\cdot)$ to show the versatility of our methodology. For Lasso and MCP penalty in our experiment, $\dot{P}_\omega(\theta) = \omega \text{sign}(\theta)$ and $\dot{P}_\omega(\theta) = (\omega - \frac{|\theta|}{a'_0})_+ \text{sign}(\theta)$ respectively, where $a'_0 = 3$.

Remark 5 We obtain uniform estimation rates for both $\hat{\beta}^\dagger$ and $\hat{a}_{rs}^\dagger(t)$ of order $O_p(\max\{(\log n/n)^{2/5}, (\log n/n^{1-\tau})^{1/2}\})$ in the Supplementary (c.f. Theorem 3 in Section C.2). Moreover, with appropriately vanishing tuning parameters and minimum signal strength, the penalized estimators achieve selection consistency: zero parametric coefficients, zero constant components, and zero varying parts are recovered with probability tending to one (c.f. Theorem 4 in Section C.2).

3.2. Tuning penalty parameters and model refinement. After obtaining $\hat{\mathbf{A}}^\dagger(t)$ and $\hat{\beta}^\dagger$, we can select variables (whether β_k can be shrunk to zero), and identify constant coefficients (whether $a_{rs}(t)$ is a zero constant function, non-zero constant function or varying function) simultaneously. To this end we need tune penalty parameters $\omega_\beta, \omega_\gamma^c$ and ω_γ^v to achieve accurate results. We thus choose the optimal tuning parameters $\omega_\beta^*, \omega_\gamma^{c,*}$ and $\omega_\gamma^{v,*}$ by a data-driven grid search method minimizing the Bayesian information criterion (BIC). Other criteria such as extended Bayesian information criterion (eBIC, Chen and Chen 2008) and generalised

information criterion (GIC, [Fan and Tang 2013](#)) can also be adopted. More specifically, we select the best tuning parameters by minimizing

$$(30) \quad \text{BIC}(\omega_\beta, \omega_\gamma^c, \omega_\gamma^v) = \log\left(\frac{\sum_{i=1}^n \sum_{j=1}^2 [y_{ij} - \langle \mathcal{H}_{ij}(t_i), \hat{\mathcal{G}}^\dagger \rangle - \mathbf{z}_i^\top \hat{\boldsymbol{\beta}}^\dagger]^2}{2n}\right) + (d_\beta + d_c + d_v L) \frac{\log(n)}{n},$$

where $\hat{\mathcal{G}}^\dagger$ and $\hat{\boldsymbol{\beta}}^\dagger$ are the solution of (23) given $\omega_\beta, \omega_\gamma^c$ and ω_γ^v , d_c and d_v are the numbers of functions estimated as nonzero constant functions and varying functions respectively, and d_β is the number of non-zero coefficients in $\hat{\boldsymbol{\beta}}^\dagger$. L is equivalent number of constant coefficients for a varying function, and exactly equal to the number of spline basis in (23).

4. Simulation study. We evaluate the finite-sample performance of the paired-eye estimation procedure under two- and three-dimensional tensor covariates. For subject i and eye $j \in \{1, 2\}$, data are generated according to (3), where $t_i \stackrel{\text{iid}}{\sim} U(0, 1)$, $\mathbf{z}_i \in \mathbb{R}^{p_0}$ has independent standard Gaussian entries, $p_0 = 4$, and $\boldsymbol{\beta} = (2, 1, -1, 0.5)^\top$. The two errors from the same subject satisfy

$$(31) \quad \begin{pmatrix} \epsilon_{i1} \\ \epsilon_{i2} \end{pmatrix} | t_i \sim N \left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \sigma^2(t_i) \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix} \right\}.$$

We consider $\rho \in \{0, 0.3, 0.6, 0.9\}$.

For both tensor dimensions, the coefficient tensor in block s is generated as

$$(32) \quad \mathcal{A}^{(s)}(t) = \sum_{r=1}^R \sqrt{\frac{rs}{RS}} \alpha(t) \mathbf{x}_r^{(s1)} \circ \dots \circ \mathbf{x}_r^{(sD)}.$$

The common shape function $\alpha(t)$ has six choices, denoted by (i)–(vi), which induce six choices for the full coefficient tensor $\mathcal{A}(t)$:

- (i) $\alpha(t) = 5(t - 0.2)^2$, (ii) $\alpha(t) = e^{-(3t-1)^2} - 0.75$, (iii) $\alpha(t) = \sin\{2\pi(t - 0.5)\}$,
- (iv) $\alpha(t) = 1.8(t - 0.2)^2 + 0.35 \left[e^{-(3t-1)^2} - 0.75 \right] + 0.20 \sin\{2\pi(t - 0.5)\}$,
- (v) $\alpha(t) = 1.10e^{-(t-0.30)^2/0.0128} - 0.95e^{-(t-0.72)^2/0.0242}$,
- (vi) $\alpha(t) = 18(t - 0.2)(t - 0.55)(t - 0.85) - 0.05$.

The variance function $\sigma^2(t)$ has four choices:

- (a) $\sigma^2(t) \equiv 1$, (b) $\sigma^2(t) = 1 + 0.3 \sin(2\pi t)$,
- (c) $\sigma^2(t) = 1 + 0.25 \cos(2\pi t) + 0.1 \sin(4\pi t)$, (d) $\sigma^2(t) = 0.5 + 0.5 \sin^2(\pi t)$.

For both designs, the s th tensor block is generated by

$$(33) \quad \mathcal{X}_{ij}^{(s)} = \sum_{r=1}^R x_{ijrs}^* \mathbf{x}_r^{(s1)} \circ \dots \circ \mathbf{x}_r^{(sD)}, \quad x_{ijrs}^* \stackrel{\text{iid}}{\sim} N(0, 1),$$

where the mode-specific factors are orthonormal columns obtained by QR decomposition of Gaussian random matrices. The two-dimensional design uses $D = 2$, $p_1 = p_2 = 60$, $p'_1 = p'_2 = 12$, $S^{(1)} = S^{(2)} = 5$, $S = 25$, $R = 4$, and $n \in \{1000, 2000\}$. The three-dimensional design uses $D = 3$, $p_1 = p_2 = p_3 = 48$, $p'_1 = p'_2 = p'_3 = 16$, $S^{(1)} = S^{(2)} = S^{(3)} = 3$, $S = 27$, $R = 6$, and $n \in \{2000, 5000\}$.

For each tensor dimension, crossing the six coefficient choices with the four variance choices gives 24 Cases. If $q = 1, \dots, 6$ indexes coefficient choices (i)–(vi) and $v = 1, \dots, 4$ indexes variance choices (a)–(d), the two-dimensional Case number is $6(v - 1) + q$, and the corresponding three-dimensional Case number is $24 + 6(v - 1) + q$. Thus, Cases 1–24 correspond to the two-dimensional design and Cases 25–48 to the three-dimensional design; sample size and ρ vary within each Case.

For each combination of Case, sample size, and ρ , we generate 30 replications. The independence estimates $\hat{\mathbf{A}}^\dagger(t)$ and $\hat{\beta}^\dagger$ are obtained by treating the two eyes as independent, whereas the proposed paired estimates $\hat{\mathbf{A}}^*(t)$ and $\hat{\beta}^*$ incorporate the estimated within-subject covariance structure. We report MIAE for $\mathcal{A}(t)$ and $\sigma^2(t)$ and MAE for β and ρ . Values are means over the 30 replications, with standard deviations in parentheses. Tables 1–4 report the two-dimensional results for Cases 1–24, and Tables 5–8 report the three-dimensional results for Cases 25–48.

Tables 1–4 and 5–8 demonstrate the benefit of the proposed paired estimation procedure across both tensor dimensions. When $\rho = 0$, the independence and paired coefficient estimates are nearly indistinguishable, showing that incorporating the paired covariance structure entails essentially no loss when the two eyes are independent. As within-subject correlation increases, the paired estimator generally yields a lower coefficient-function MIAE, with the largest improvements occurring at $\rho = 0.9$. Estimation errors for the coefficient, parametric, and covariance components also generally decrease as the sample size increases. Although the gains for β are smaller and vary by configuration, the numerical results show that the proposed procedure remains stable and effectively exploits paired-eye dependence when it is present.

Figure 2 provides complementary diagnostics: the paired coefficient-function curves closely track the truth and generally improve upon the independence curves, while the estimated variance-function curves recover distinct heteroscedastic patterns. Together, the tabular and graphical results show that the proposed method accurately recovers both the regression signal and the within-subject covariance structure.

TABLE 1
Estimation errors ($\times 10^{-2}$) for Cases 1–6.

Case	n	ρ	MIAE		MAE		MIAE	MAE
			$\hat{\mathbf{A}}^\dagger(t)$	$\hat{\mathbf{A}}^*(t)$	$\hat{\beta}^\dagger$	$\hat{\beta}^*$	$\hat{\sigma}^2(t)$	$\hat{\rho}$
Case 1	1000	0.0	5.06 (± 0.17)	5.08 (± 0.18)	2.09 (± 0.78)	2.12 (± 0.77)	30.32 (± 2.66)	2.81 (± 1.91)
		0.3	5.05 (± 0.20)	4.94 (± 0.18)	2.44 (± 1.01)	2.49 (± 1.07)	30.23 (± 3.48)	7.48 (± 3.06)
		0.6	5.06 (± 0.24)	4.53 (± 0.17)	2.66 (± 1.10)	2.69 (± 1.13)	30.19 (± 3.71)	15.09 (± 2.55)
		0.9	5.06 (± 0.27)	3.57 (± 0.18)	2.88 (± 1.13)	2.84 (± 1.15)	30.14 (± 3.99)	22.70 (± 1.92)
	2000	0.0	3.29 (± 0.10)	3.30 (± 0.10)	1.25 (± 0.43)	1.24 (± 0.44)	13.31 (± 1.88)	1.37 (± 0.80)
		0.3	3.32 (± 0.11)	3.22 (± 0.11)	1.39 (± 0.59)	1.36 (± 0.61)	13.24 (± 1.82)	4.12 (± 1.64)
		0.6	3.33 (± 0.11)	2.89 (± 0.10)	1.53 (± 0.65)	1.47 (± 0.67)	13.15 (± 2.04)	8.48 (± 1.41)
		0.9	3.33 (± 0.11)	2.16 (± 0.07)	1.68 (± 0.70)	1.56 (± 0.72)	13.07 (± 2.32)	12.92 (± 1.03)
Case 2	1000	0.0	5.08 (± 0.21)	5.10 (± 0.21)	2.09 (± 0.79)	2.12 (± 0.78)	30.33 (± 2.72)	2.94 (± 2.11)
		0.3	5.03 (± 0.18)	4.92 (± 0.16)	2.44 (± 0.98)	2.51 (± 1.03)	30.28 (± 3.56)	6.87 (± 2.92)
		0.6	5.03 (± 0.20)	4.50 (± 0.16)	2.66 (± 1.08)	2.71 (± 1.11)	30.25 (± 3.78)	14.46 (± 2.31)
		0.9	5.02 (± 0.22)	3.52 (± 0.15)	2.89 (± 1.12)	2.84 (± 1.14)	30.20 (± 4.05)	22.09 (± 1.48)
	2000	0.0	3.28 (± 0.08)	3.29 (± 0.08)	1.21 (± 0.46)	1.21 (± 0.46)	13.43 (± 1.89)	1.56 (± 0.96)
		0.3	3.29 (± 0.12)	3.19 (± 0.11)	1.39 (± 0.59)	1.37 (± 0.63)	13.34 (± 1.78)	3.87 (± 1.59)
		0.6	3.29 (± 0.12)	2.86 (± 0.10)	1.54 (± 0.64)	1.48 (± 0.70)	13.24 (± 2.03)	8.22 (± 1.31)
		0.9	3.29 (± 0.11)	2.09 (± 0.08)	1.69 (± 0.69)	1.60 (± 0.74)	13.12 (± 2.34)	12.64 (± 0.98)
Case 3	1000	0.0	6.49 (± 0.22)	6.60 (± 0.23)	2.21 (± 0.76)	2.27 (± 0.76)	17.16 (± 2.64)	2.85 (± 2.21)
		0.3	6.52 (± 0.18)	6.56 (± 0.17)	2.46 (± 1.11)	2.53 (± 1.09)	17.08 (± 3.21)	11.00 (± 3.08)
		0.6	6.53 (± 0.19)	6.29 (± 0.16)	2.70 (± 1.18)	2.73 (± 1.16)	17.15 (± 3.26)	22.32 (± 2.77)
		0.9	6.54 (± 0.22)	5.75 (± 0.15)	2.95 (± 1.21)	2.89 (± 1.22)	17.24 (± 3.35)	33.66 (± 2.46)
	2000	0.0	5.04 (± 0.11)	5.14 (± 0.12)	1.49 (± 0.56)	1.48 (± 0.58)	9.82 (± 1.17)	1.33 (± 0.98)
		0.3	5.05 (± 0.12)	5.09 (± 0.12)	1.41 (± 0.59)	1.38 (± 0.61)	10.29 (± 1.30)	8.32 (± 1.90)
		0.6	5.06 (± 0.13)	4.89 (± 0.11)	1.55 (± 0.65)	1.49 (± 0.66)	10.41 (± 1.36)	16.81 (± 1.83)
		0.9	5.06 (± 0.13)	4.50 (± 0.09)	1.67 (± 0.70)	1.58 (± 0.70)	10.57 (± 1.48)	25.32 (± 1.67)
Case 4	1000	0.0	4.87 (± 0.19)	4.88 (± 0.20)	2.09 (± 0.76)	2.13 (± 0.75)	31.77 (± 2.69)	2.85 (± 1.97)
		0.3	4.85 (± 0.18)	4.73 (± 0.16)	2.43 (± 1.01)	2.47 (± 1.07)	31.68 (± 3.47)	6.83 (± 2.97)
		0.6	4.85 (± 0.21)	4.28 (± 0.15)	2.65 (± 1.10)	2.67 (± 1.14)	31.64 (± 3.69)	13.98 (± 2.37)
		0.9	4.86 (± 0.24)	3.20 (± 0.16)	2.88 (± 1.13)	2.82 (± 1.15)	31.59 (± 3.97)	21.14 (± 1.64)
	2000	0.0	3.02 (± 0.09)	3.02 (± 0.09)	1.26 (± 0.45)	1.26 (± 0.45)	15.12 (± 1.87)	1.44 (± 0.78)
		0.3	3.03 (± 0.09)	2.91 (± 0.09)	1.36 (± 0.58)	1.33 (± 0.60)	15.04 (± 1.85)	3.57 (± 1.45)
		0.6	3.04 (± 0.09)	2.54 (± 0.08)	1.51 (± 0.64)	1.45 (± 0.66)	14.95 (± 2.06)	7.33 (± 1.30)
		0.9	3.04 (± 0.09)	1.63 (± 0.06)	1.65 (± 0.69)	1.54 (± 0.71)	14.85 (± 2.35)	11.21 (± 0.88)
Case 5	1000	0.0	7.88 (± 0.22)	8.19 (± 0.24)	2.56 (± 1.05)	2.49 (± 1.00)	28.32 (± 2.37)	3.50 (± 2.04)
		0.3	7.84 (± 0.26)	8.09 (± 0.26)	2.63 (± 0.83)	2.71 (± 0.96)	28.28 (± 2.54)	14.55 (± 3.56)
		0.6	7.85 (± 0.27)	7.89 (± 0.26)	2.85 (± 0.88)	2.93 (± 1.03)	28.21 (± 2.58)	29.92 (± 3.58)
		0.9	7.85 (± 0.30)	7.48 (± 0.28)	3.08 (± 0.93)	3.13 (± 1.11)	28.19 (± 2.66)	45.30 (± 3.57)
	2000	0.0	6.46 (± 0.14)	6.77 (± 0.14)	1.59 (± 0.55)	1.53 (± 0.50)	38.31 (± 2.77)	2.00 (± 1.43)
		0.3	6.48 (± 0.13)	6.73 (± 0.13)	1.66 (± 0.63)	1.61 (± 0.55)	37.98 (± 2.08)	13.04 (± 2.74)
		0.6	6.49 (± 0.13)	6.57 (± 0.13)	1.79 (± 0.67)	1.74 (± 0.59)	38.02 (± 2.04)	26.38 (± 2.59)
		0.9	6.48 (± 0.13)	6.24 (± 0.13)	1.93 (± 0.71)	1.85 (± 0.61)	38.11 (± 2.09)	39.79 (± 2.32)
Case 6	1000	0.0	6.18 (± 0.21)	6.26 (± 0.21)	2.10 (± 0.84)	2.15 (± 0.81)	20.74 (± 2.87)	2.91 (± 2.08)
		0.3	6.13 (± 0.21)	6.12 (± 0.20)	2.62 (± 1.01)	2.64 (± 1.05)	20.63 (± 3.74)	9.38 (± 3.13)
		0.6	6.12 (± 0.23)	5.81 (± 0.20)	2.83 (± 1.12)	2.82 (± 1.13)	20.62 (± 3.94)	19.66 (± 2.69)
		0.9	6.11 (± 0.25)	5.18 (± 0.20)	3.04 (± 1.19)	2.96 (± 1.18)	20.62 (± 4.18)	29.96 (± 2.07)
	2000	0.0	4.59 (± 0.10)	4.65 (± 0.10)	1.17 (± 0.45)	1.17 (± 0.45)	6.90 (± 1.09)	1.67 (± 1.00)
		0.3	4.61 (± 0.13)	4.61 (± 0.12)	1.55 (± 0.64)	1.54 (± 0.67)	7.11 (± 1.22)	6.86 (± 1.85)
		0.6	4.61 (± 0.13)	4.40 (± 0.11)	1.71 (± 0.69)	1.66 (± 0.74)	7.16 (± 1.31)	14.36 (± 1.69)
		0.9	4.60 (± 0.14)	3.95 (± 0.08)	1.86 (± 0.73)	1.79 (± 0.78)	7.26 (± 1.44)	21.94 (± 1.52)

TABLE 2
Estimation errors ($\times 10^{-2}$) for Cases 7–12.

Case	n	ρ	MIAE		MAE		MIAE	MAE
			$\hat{\mathbf{A}}^\dagger(t)$	$\hat{\mathbf{A}}^*(t)$	$\hat{\beta}^\dagger$	$\hat{\beta}^*$	$\hat{\sigma}^2(t)$	$\hat{\rho}$
Case 7	1000	0.0	5.05 ($\pm$ 0.18)	5.05 ($\pm$ 0.18)	2.12 ($\pm$ 0.80)	2.05 ($\pm$ 0.75)	30.43 ($\pm$ 2.75)	2.93 ($\pm$ 1.94)
		0.3	5.03 ($\pm$ 0.20)	4.92 ($\pm$ 0.18)	2.41 ($\pm$ 1.03)	2.45 ($\pm$ 1.04)	30.27 ($\pm$ 3.47)	7.63 ($\pm$ 3.19)
		0.6	5.03 ($\pm$ 0.23)	4.51 ($\pm$ 0.17)	2.63 ($\pm$ 1.10)	2.65 ($\pm$ 1.10)	30.26 ($\pm$ 3.76)	15.23 ($\pm$ 2.67)
		0.9	5.04 ($\pm$ 0.27)	3.56 ($\pm$ 0.18)	2.86 ($\pm$ 1.14)	2.79 ($\pm$ 1.12)	30.23 ($\pm$ 4.12)	22.80 ($\pm$ 2.02)
	2000	0.0	3.29 ($\pm$ 0.10)	3.29 ($\pm$ 0.10)	1.24 ($\pm$ 0.46)	1.21 ($\pm$ 0.42)	13.47 ($\pm$ 1.92)	1.60 ($\pm$ 0.90)
		0.3	3.31 ($\pm$ 0.11)	3.20 ($\pm$ 0.11)	1.39 ($\pm$ 0.61)	1.34 ($\pm$ 0.58)	13.41 ($\pm$ 1.93)	4.31 ($\pm$ 1.55)
		0.6	3.32 ($\pm$ 0.11)	2.88 ($\pm$ 0.10)	1.53 ($\pm$ 0.67)	1.44 ($\pm$ 0.65)	13.46 ($\pm$ 2.10)	8.63 ($\pm$ 1.31)
		0.9	3.32 ($\pm$ 0.11)	2.15 ($\pm$ 0.07)	1.67 ($\pm$ 0.72)	1.53 ($\pm$ 0.69)	13.57 ($\pm$ 2.25)	12.97 ($\pm$ 0.98)
Case 8	1000	0.0	5.05 ($\pm$ 0.21)	5.07 ($\pm$ 0.21)	2.11 ($\pm$ 0.82)	2.06 ($\pm$ 0.75)	30.43 ($\pm$ 2.82)	3.14 ($\pm$ 2.03)
		0.3	4.99 ($\pm$ 0.18)	4.89 ($\pm$ 0.17)	2.42 ($\pm$ 1.01)	2.44 ($\pm$ 1.02)	30.32 ($\pm$ 3.54)	6.97 ($\pm$ 3.07)
		0.6	4.99 ($\pm$ 0.20)	4.47 ($\pm$ 0.16)	2.66 ($\pm$ 1.09)	2.63 ($\pm$ 1.11)	30.32 ($\pm$ 3.83)	14.55 ($\pm$ 2.42)
		0.9	4.99 ($\pm$ 0.22)	3.49 ($\pm$ 0.15)	2.88 ($\pm$ 1.15)	2.78 ($\pm$ 1.12)	30.30 ($\pm$ 4.18)	22.12 ($\pm$ 1.51)
	2000	0.0	3.26 ($\pm$ 0.08)	3.27 ($\pm$ 0.08)	1.19 ($\pm$ 0.46)	1.18 ($\pm$ 0.44)	13.55 ($\pm$ 1.99)	1.77 ($\pm$ 1.12)
		0.3	3.26 ($\pm$ 0.12)	3.17 ($\pm$ 0.12)	1.38 ($\pm$ 0.62)	1.35 ($\pm$ 0.58)	13.41 ($\pm$ 1.94)	4.06 ($\pm$ 1.53)
		0.6	3.26 ($\pm$ 0.12)	2.84 ($\pm$ 0.10)	1.54 ($\pm$ 0.67)	1.46 ($\pm$ 0.65)	13.41 ($\pm$ 2.12)	8.36 ($\pm$ 1.28)
		0.9	3.26 ($\pm$ 0.11)	2.07 ($\pm$ 0.08)	1.69 ($\pm$ 0.71)	1.56 ($\pm$ 0.70)	13.45 ($\pm$ 2.32)	12.69 ($\pm$ 1.00)
Case 9	1000	0.0	6.48 ($\pm$ 0.22)	6.59 ($\pm$ 0.23)	2.22 ($\pm$ 0.79)	2.21 ($\pm$ 0.75)	19.24 ($\pm$ 2.34)	2.95 ($\pm$ 2.18)
		0.3	6.51 ($\pm$ 0.18)	6.54 ($\pm$ 0.17)	2.43 ($\pm$ 1.08)	2.49 ($\pm$ 1.08)	19.03 ($\pm$ 2.81)	11.20 ($\pm$ 3.26)
		0.6	6.52 ($\pm$ 0.20)	6.28 ($\pm$ 0.17)	2.67 ($\pm$ 1.15)	2.69 ($\pm$ 1.15)	19.03 ($\pm$ 3.09)	22.52 ($\pm$ 2.99)
		0.9	6.53 ($\pm$ 0.22)	5.74 ($\pm$ 0.16)	2.91 ($\pm$ 1.19)	2.86 ($\pm$ 1.20)	19.06 ($\pm$ 3.45)	33.85 ($\pm$ 2.68)
	2000	0.0	5.04 ($\pm$ 0.11)	5.14 ($\pm$ 0.12)	1.52 ($\pm$ 0.55)	1.47 ($\pm$ 0.56)	10.30 ($\pm$ 1.13)	1.39 ($\pm$ 1.04)
		0.3	5.05 ($\pm$ 0.12)	5.08 ($\pm$ 0.12)	1.41 ($\pm$ 0.58)	1.35 ($\pm$ 0.61)	10.84 ($\pm$ 1.29)	8.51 ($\pm$ 1.89)
		0.6	5.06 ($\pm$ 0.13)	4.89 ($\pm$ 0.11)	1.54 ($\pm$ 0.63)	1.46 ($\pm$ 0.66)	10.99 ($\pm$ 1.33)	16.96 ($\pm$ 1.81)
		0.9	5.06 ($\pm$ 0.13)	4.49 ($\pm$ 0.09)	1.67 ($\pm$ 0.68)	1.56 ($\pm$ 0.70)	11.22 ($\pm$ 1.37)	25.42 ($\pm$ 1.66)
Case 10	1000	0.0	4.85 ($\pm$ 0.20)	4.86 ($\pm$ 0.20)	2.12 ($\pm$ 0.78)	2.05 ($\pm$ 0.71)	31.88 ($\pm$ 2.79)	2.95 ($\pm$ 2.01)
		0.3	4.82 ($\pm$ 0.18)	4.70 ($\pm$ 0.16)	2.39 ($\pm$ 1.03)	2.42 ($\pm$ 1.05)	31.72 ($\pm$ 3.45)	6.98 ($\pm$ 3.13)
		0.6	4.82 ($\pm$ 0.21)	4.26 ($\pm$ 0.15)	2.62 ($\pm$ 1.10)	2.61 ($\pm$ 1.12)	31.71 ($\pm$ 3.75)	14.11 ($\pm$ 2.52)
		0.9	4.82 ($\pm$ 0.24)	3.18 ($\pm$ 0.15)	2.85 ($\pm$ 1.14)	2.77 ($\pm$ 1.13)	31.69 ($\pm$ 4.10)	21.23 ($\pm$ 1.76)
	2000	0.0	3.00 ($\pm$ 0.09)	3.00 ($\pm$ 0.09)	1.25 ($\pm$ 0.47)	1.23 ($\pm$ 0.43)	15.24 ($\pm$ 1.94)	1.65 ($\pm$ 0.92)
		0.3	3.01 ($\pm$ 0.09)	2.89 ($\pm$ 0.09)	1.36 ($\pm$ 0.59)	1.31 ($\pm$ 0.57)	15.15 ($\pm$ 1.94)	3.71 ($\pm$ 1.47)
		0.6	3.02 ($\pm$ 0.09)	2.52 ($\pm$ 0.08)	1.51 ($\pm$ 0.65)	1.42 ($\pm$ 0.64)	15.14 ($\pm$ 2.11)	7.47 ($\pm$ 1.21)
		0.9	3.02 ($\pm$ 0.09)	1.62 ($\pm$ 0.06)	1.65 ($\pm$ 0.70)	1.51 ($\pm$ 0.68)	15.15 ($\pm$ 2.33)	11.25 ($\pm$ 0.85)
Case 11	1000	0.0	7.85 ($\pm$ 0.23)	8.15 ($\pm$ 0.25)	2.57 ($\pm$ 1.07)	2.44 ($\pm$ 0.94)	26.70 ($\pm$ 2.27)	3.61 ($\pm$ 2.24)
		0.3	7.81 ($\pm$ 0.25)	8.05 ($\pm$ 0.25)	2.63 ($\pm$ 0.82)	2.60 ($\pm$ 0.94)	26.69 ($\pm$ 2.51)	14.59 ($\pm$ 3.55)
		0.6	7.81 ($\pm$ 0.26)	7.85 ($\pm$ 0.25)	2.85 ($\pm$ 0.86)	2.82 ($\pm$ 1.02)	26.63 ($\pm$ 2.58)	29.96 ($\pm$ 3.56)
		0.9	7.81 ($\pm$ 0.28)	7.45 ($\pm$ 0.27)	3.08 ($\pm$ 0.90)	3.02 ($\pm$ 1.08)	26.58 ($\pm$ 2.69)	45.32 ($\pm$ 3.51)
	2000	0.0	6.44 ($\pm$ 0.13)	6.74 ($\pm$ 0.14)	1.58 ($\pm$ 0.53)	1.48 ($\pm$ 0.47)	38.26 ($\pm$ 2.98)	2.13 ($\pm$ 1.50)
		0.3	6.46 ($\pm$ 0.13)	6.70 ($\pm$ 0.13)	1.67 ($\pm$ 0.63)	1.57 ($\pm$ 0.56)	37.88 ($\pm$ 2.37)	13.22 ($\pm$ 2.83)
		0.6	6.46 ($\pm$ 0.13)	6.54 ($\pm$ 0.13)	1.80 ($\pm$ 0.67)	1.69 ($\pm$ 0.60)	37.88 ($\pm$ 2.32)	26.55 ($\pm$ 2.68)
		0.9	6.46 ($\pm$ 0.13)	6.22 ($\pm$ 0.13)	1.93 ($\pm$ 0.71)	1.80 ($\pm$ 0.62)	37.90 ($\pm$ 2.35)	39.91 ($\pm$ 2.40)
Case 12	1000	0.0	6.16 ($\pm$ 0.21)	6.23 ($\pm$ 0.22)	2.11 ($\pm$ 0.87)	2.11 ($\pm$ 0.79)	20.88 ($\pm$ 2.96)	3.10 ($\pm$ 2.08)
		0.3	6.10 ($\pm$ 0.21)	6.09 ($\pm$ 0.20)	2.62 ($\pm$ 1.05)	2.57 ($\pm$ 1.01)	20.76 ($\pm$ 3.59)	9.42 ($\pm$ 3.30)
		0.6	6.09 ($\pm$ 0.23)	5.78 ($\pm$ 0.20)	2.83 ($\pm$ 1.15)	2.75 ($\pm$ 1.09)	20.79 ($\pm$ 3.83)	19.70 ($\pm$ 2.82)
		0.9	6.08 ($\pm$ 0.24)	5.16 ($\pm$ 0.20)	3.04 ($\pm$ 1.23)	2.88 ($\pm$ 1.15)	20.80 ($\pm$ 4.15)	29.97 ($\pm$ 2.13)
	2000	0.0	4.58 ($\pm$ 0.10)	4.64 ($\pm$ 0.10)	1.16 ($\pm$ 0.45)	1.16 ($\pm$ 0.44)	6.98 ($\pm$ 1.29)	1.80 ($\pm$ 1.19)
		0.3	4.58 ($\pm$ 0.13)	4.59 ($\pm$ 0.12)	1.53 ($\pm$ 0.69)	1.52 ($\pm$ 0.64)	7.08 ($\pm$ 1.29)	7.02 ($\pm$ 1.88)
		0.6	4.59 ($\pm$ 0.13)	4.38 ($\pm$ 0.11)	1.68 ($\pm$ 0.75)	1.64 ($\pm$ 0.70)	7.21 ($\pm$ 1.49)	14.49 ($\pm$ 1.74)
		0.9	4.58 ($\pm$ 0.14)	3.94 ($\pm$ 0.08)	1.84 ($\pm$ 0.79)	1.75 ($\pm$ 0.74)	7.38 ($\pm$ 1.73)	22.00 ($\pm$ 1.61)

TABLE 3
Estimation errors ($\times 10^{-2}$) for Cases 13–18.

Case	n	ρ	MIAE		MAE		MIAE	MAE
			$\hat{\mathbf{A}}^\dagger(t)$	$\hat{\mathbf{A}}^*(t)$	$\hat{\beta}^\dagger$	$\hat{\beta}^*$	$\hat{\sigma}^2(t)$	$\hat{\rho}$
Case 13	1000	0.0	4.28 ($\pm$ 0.14)	4.29 ($\pm$ 0.14)	1.82 ($\pm$ 0.68)	1.78 ($\pm$ 0.66)	20.98 ($\pm$ 2.10)	2.79 ($\pm$ 1.96)
		0.3	4.28 ($\pm$ 0.18)	4.19 ($\pm$ 0.16)	2.08 ($\pm$ 0.85)	2.12 ($\pm$ 0.89)	20.91 ($\pm$ 2.62)	7.39 ($\pm$ 3.17)
		0.6	4.28 ($\pm$ 0.20)	3.84 ($\pm$ 0.15)	2.27 ($\pm$ 0.92)	2.29 ($\pm$ 0.94)	20.89 ($\pm$ 2.80)	14.77 ($\pm$ 2.63)
		0.9	4.28 ($\pm$ 0.23)	3.06 ($\pm$ 0.15)	2.45 ($\pm$ 0.96)	2.40 ($\pm$ 0.96)	20.85 ($\pm$ 3.03)	22.15 ($\pm$ 1.96)
	2000	0.0	2.85 ($\pm$ 0.09)	2.86 ($\pm$ 0.09)	1.09 ($\pm$ 0.38)	1.04 ($\pm$ 0.38)	8.56 ($\pm$ 1.44)	1.51 ($\pm$ 0.91)
		0.3	2.87 ($\pm$ 0.09)	2.79 ($\pm$ 0.09)	1.23 ($\pm$ 0.52)	1.13 ($\pm$ 0.51)	8.47 ($\pm$ 1.43)	4.17 ($\pm$ 1.73)
		0.6	2.88 ($\pm$ 0.09)	2.53 ($\pm$ 0.09)	1.35 ($\pm$ 0.57)	1.23 ($\pm$ 0.56)	8.41 ($\pm$ 1.58)	8.51 ($\pm$ 1.42)
		0.9	2.88 ($\pm$ 0.10)	1.96 ($\pm$ 0.06)	1.48 ($\pm$ 0.61)	1.31 ($\pm$ 0.60)	8.37 ($\pm$ 1.77)	12.91 ($\pm$ 1.01)
Case 14	1000	0.0	4.29 ($\pm$ 0.17)	4.31 ($\pm$ 0.17)	1.83 ($\pm$ 0.69)	1.77 ($\pm$ 0.67)	21.00 ($\pm$ 2.15)	3.03 ($\pm$ 2.04)
		0.3	4.24 ($\pm$ 0.15)	4.16 ($\pm$ 0.14)	2.08 ($\pm$ 0.83)	2.13 ($\pm$ 0.87)	20.97 ($\pm$ 2.70)	6.68 ($\pm$ 2.98)
		0.6	4.24 ($\pm$ 0.17)	3.81 ($\pm$ 0.13)	2.27 ($\pm$ 0.89)	2.30 ($\pm$ 0.93)	20.95 ($\pm$ 2.87)	14.04 ($\pm$ 2.36)
		0.9	4.23 ($\pm$ 0.18)	3.00 ($\pm$ 0.13)	2.48 ($\pm$ 0.92)	2.41 ($\pm$ 0.97)	20.91 ($\pm$ 3.09)	21.43 ($\pm$ 1.49)
	2000	0.0	2.83 ($\pm$ 0.07)	2.84 ($\pm$ 0.07)	1.04 ($\pm$ 0.40)	1.01 ($\pm$ 0.39)	8.68 ($\pm$ 1.45)	1.75 ($\pm$ 1.15)
		0.3	2.83 ($\pm$ 0.10)	2.76 ($\pm$ 0.10)	1.22 ($\pm$ 0.53)	1.15 ($\pm$ 0.51)	8.56 ($\pm$ 1.41)	3.87 ($\pm$ 1.72)
		0.6	2.83 ($\pm$ 0.10)	2.49 ($\pm$ 0.08)	1.36 ($\pm$ 0.56)	1.25 ($\pm$ 0.58)	8.49 ($\pm$ 1.58)	8.19 ($\pm$ 1.38)
		0.9	2.83 ($\pm$ 0.10)	1.87 ($\pm$ 0.06)	1.49 ($\pm$ 0.59)	1.34 ($\pm$ 0.61)	8.42 ($\pm$ 1.81)	12.56 ($\pm$ 0.99)
Case 15	1000	0.0	5.84 ($\pm$ 0.19)	5.96 ($\pm$ 0.20)	1.94 ($\pm$ 0.68)	1.95 ($\pm$ 0.68)	9.43 ($\pm$ 1.33)	3.00 ($\pm$ 2.14)
		0.3	5.88 ($\pm$ 0.15)	5.94 ($\pm$ 0.15)	2.12 ($\pm$ 0.98)	2.15 ($\pm$ 0.90)	9.57 ($\pm$ 1.61)	11.74 ($\pm$ 3.09)
		0.6	5.88 ($\pm$ 0.16)	5.74 ($\pm$ 0.14)	2.32 ($\pm$ 1.05)	2.32 ($\pm$ 0.95)	9.61 ($\pm$ 1.60)	23.73 ($\pm$ 2.84)
		0.9	5.89 ($\pm$ 0.17)	5.34 ($\pm$ 0.13)	2.54 ($\pm$ 1.06)	2.45 ($\pm$ 1.00)	9.69 ($\pm$ 1.68)	35.74 ($\pm$ 2.62)
	2000	0.0	4.72 ($\pm$ 0.10)	4.84 ($\pm$ 0.11)	1.36 ($\pm$ 0.51)	1.29 ($\pm$ 0.51)	10.46 ($\pm$ 0.95)	1.38 ($\pm$ 1.05)
		0.3	4.72 ($\pm$ 0.11)	4.80 ($\pm$ 0.10)	1.28 ($\pm$ 0.50)	1.17 ($\pm$ 0.52)	10.69 ($\pm$ 1.10)	9.40 ($\pm$ 2.06)
		0.6	4.73 ($\pm$ 0.12)	4.65 ($\pm$ 0.10)	1.39 ($\pm$ 0.55)	1.26 ($\pm$ 0.56)	10.78 ($\pm$ 1.15)	18.86 ($\pm$ 1.97)
		0.9	4.73 ($\pm$ 0.12)	4.37 ($\pm$ 0.08)	1.50 ($\pm$ 0.59)	1.34 ($\pm$ 0.60)	10.92 ($\pm$ 1.25)	28.35 ($\pm$ 1.83)
Case 16	1000	0.0	4.07 ($\pm$ 0.15)	4.07 ($\pm$ 0.16)	1.81 ($\pm$ 0.67)	1.79 ($\pm$ 0.64)	22.43 ($\pm$ 2.13)	2.89 ($\pm$ 1.95)
		0.3	4.05 ($\pm$ 0.15)	3.95 ($\pm$ 0.14)	2.07 ($\pm$ 0.86)	2.09 ($\pm$ 0.89)	22.37 ($\pm$ 2.61)	6.57 ($\pm$ 3.04)
		0.6	4.06 ($\pm$ 0.18)	3.57 ($\pm$ 0.13)	2.26 ($\pm$ 0.93)	2.27 ($\pm$ 0.94)	22.34 ($\pm$ 2.78)	13.34 ($\pm$ 2.41)
		0.9	4.05 ($\pm$ 0.20)	2.65 ($\pm$ 0.13)	2.46 ($\pm$ 0.96)	2.38 ($\pm$ 0.96)	22.31 ($\pm$ 3.01)	20.13 ($\pm$ 1.62)
	2000	0.0	2.55 ($\pm$ 0.08)	2.55 ($\pm$ 0.08)	1.10 ($\pm$ 0.40)	1.05 ($\pm$ 0.38)	10.37 ($\pm$ 1.44)	1.56 ($\pm$ 0.90)
		0.3	2.56 ($\pm$ 0.08)	2.45 ($\pm$ 0.08)	1.19 ($\pm$ 0.50)	1.11 ($\pm$ 0.50)	10.27 ($\pm$ 1.46)	3.45 ($\pm$ 1.54)
		0.6	2.56 ($\pm$ 0.08)	2.15 ($\pm$ 0.07)	1.33 ($\pm$ 0.55)	1.21 ($\pm$ 0.55)	10.21 ($\pm$ 1.61)	7.01 ($\pm$ 1.30)
		0.9	2.56 ($\pm$ 0.08)	1.39 ($\pm$ 0.05)	1.45 ($\pm$ 0.59)	1.28 ($\pm$ 0.59)	10.14 ($\pm$ 1.82)	10.67 ($\pm$ 0.83)
Case 17	1000	0.0	7.99 ($\pm$ 0.23)	8.31 ($\pm$ 0.25)	2.56 ($\pm$ 1.06)	2.50 ($\pm$ 1.00)	32.41 ($\pm$ 2.33)	3.50 ($\pm$ 2.03)
		0.3	7.96 ($\pm$ 0.26)	8.21 ($\pm$ 0.26)	2.64 ($\pm$ 0.83)	2.69 ($\pm$ 0.96)	32.40 ($\pm$ 2.47)	14.82 ($\pm$ 3.62)
		0.6	7.96 ($\pm$ 0.28)	8.00 ($\pm$ 0.27)	2.87 ($\pm$ 0.90)	2.92 ($\pm$ 1.03)	32.35 ($\pm$ 2.51)	30.51 ($\pm$ 3.66)
		0.9	7.97 ($\pm$ 0.30)	7.60 ($\pm$ 0.29)	3.09 ($\pm$ 0.96)	3.12 ($\pm$ 1.09)	32.35 ($\pm$ 2.59)	46.21 ($\pm$ 3.67)
	2000	0.0	6.52 ($\pm$ 0.13)	6.83 ($\pm$ 0.14)	1.58 ($\pm$ 0.56)	1.53 ($\pm$ 0.49)	40.76 ($\pm$ 2.52)	1.90 ($\pm$ 1.35)
		0.3	6.55 ($\pm$ 0.13)	6.80 ($\pm$ 0.13)	1.67 ($\pm$ 0.62)	1.61 ($\pm$ 0.58)	40.52 ($\pm$ 1.85)	13.24 ($\pm$ 2.74)
		0.6	6.55 ($\pm$ 0.13)	6.63 ($\pm$ 0.13)	1.80 ($\pm$ 0.65)	1.74 ($\pm$ 0.62)	40.56 ($\pm$ 1.81)	26.77 ($\pm$ 2.60)
		0.9	6.55 ($\pm$ 0.14)	6.29 ($\pm$ 0.13)	1.93 ($\pm$ 0.69)	1.85 ($\pm$ 0.64)	40.62 ($\pm$ 1.89)	40.36 ($\pm$ 2.34)
Case 18	1000	0.0	6.26 ($\pm$ 0.22)	6.34 ($\pm$ 0.22)	2.08 ($\pm$ 0.86)	2.14 ($\pm$ 0.78)	22.13 ($\pm$ 2.88)	3.02 ($\pm$ 2.21)
		0.3	6.20 ($\pm$ 0.21)	6.19 ($\pm$ 0.20)	2.64 ($\pm$ 1.05)	2.55 ($\pm$ 1.01)	21.97 ($\pm$ 3.67)	9.59 ($\pm$ 3.29)
		0.6	6.19 ($\pm$ 0.23)	5.88 ($\pm$ 0.20)	2.86 ($\pm$ 1.15)	2.73 ($\pm$ 1.10)	21.94 ($\pm$ 3.87)	20.22 ($\pm$ 2.83)
		0.9	6.18 ($\pm$ 0.26)	5.26 ($\pm$ 0.21)	3.07 ($\pm$ 1.24)	2.88 ($\pm$ 1.14)	21.92 ($\pm$ 4.10)	30.87 ($\pm$ 2.19)
	2000	0.0	4.61 ($\pm$ 0.10)	4.68 ($\pm$ 0.10)	1.18 ($\pm$ 0.46)	1.15 ($\pm$ 0.44)	7.91 ($\pm$ 1.13)	1.58 ($\pm$ 1.01)
		0.3	4.64 ($\pm$ 0.14)	4.64 ($\pm$ 0.12)	1.54 ($\pm$ 0.64)	1.50 ($\pm$ 0.68)	8.20 ($\pm$ 1.19)	7.03 ($\pm$ 1.92)
		0.6	4.64 ($\pm$ 0.14)	4.42 ($\pm$ 0.11)	1.69 ($\pm$ 0.69)	1.62 ($\pm$ 0.74)	8.24 ($\pm$ 1.26)	14.72 ($\pm$ 1.77)
		0.9	4.63 ($\pm$ 0.14)	3.97 ($\pm$ 0.09)	1.84 ($\pm$ 0.73)	1.74 ($\pm$ 0.78)	8.31 ($\pm$ 1.40)	22.50 ($\pm$ 1.60)

TABLE 4
Estimation errors ($\times 10^{-2}$) for Cases 19–24.

Case	n	ρ	MIAE		MAE		MIAE	MAE
			$\hat{\mathbf{A}}^\dagger(t)$	$\hat{\mathbf{A}}^*(t)$	$\hat{\beta}^\dagger$	$\hat{\beta}^*$	$\hat{\sigma}^2(t)$	$\hat{\rho}$
Case 19	1000	0.0	5.17 (± 0.18)	5.18 (± 0.19)	2.08 (± 0.79)	2.09 (± 0.75)	31.65 (± 2.68)	2.92 (± 2.04)
		0.3	5.15 (± 0.21)	5.03 (± 0.18)	2.45 (± 1.03)	2.44 (± 1.05)	31.52 (± 3.55)	7.70 (± 3.10)
		0.6	5.16 (± 0.24)	4.62 (± 0.17)	2.68 (± 1.10)	2.64 (± 1.11)	31.47 (± 3.79)	15.65 (± 2.60)
		0.9	5.16 (± 0.28)	3.67 (± 0.19)	2.90 (± 1.16)	2.79 (± 1.12)	31.40 (± 4.05)	23.60 (± 2.02)
	2000	0.0	3.35 (± 0.10)	3.35 (± 0.10)	1.25 (± 0.43)	1.23 (± 0.43)	14.39 (± 1.76)	1.31 (± 0.81)
		0.3	3.38 (± 0.11)	3.27 (± 0.11)	1.38 (± 0.59)	1.35 (± 0.60)	14.46 (± 1.71)	4.27 (± 1.69)
		0.6	3.39 (± 0.11)	2.94 (± 0.10)	1.53 (± 0.64)	1.46 (± 0.66)	14.38 (± 1.97)	8.81 (± 1.48)
		0.9	3.39 (± 0.11)	2.19 (± 0.07)	1.67 (± 0.69)	1.55 (± 0.70)	14.32 (± 2.29)	13.44 (± 1.10)
Case 20	1000	0.0	5.19 (± 0.22)	5.20 (± 0.22)	2.08 (± 0.80)	2.11 (± 0.76)	31.67 (± 2.75)	3.07 (± 2.20)
		0.3	5.13 (± 0.18)	5.02 (± 0.17)	2.46 (± 1.00)	2.45 (± 1.01)	31.56 (± 3.61)	7.11 (± 3.01)
		0.6	5.13 (± 0.20)	4.60 (± 0.16)	2.69 (± 1.10)	2.65 (± 1.10)	31.52 (± 3.83)	15.04 (± 2.38)
		0.9	5.13 (± 0.23)	3.62 (± 0.16)	2.92 (± 1.16)	2.80 (± 1.11)	31.47 (± 4.09)	23.00 (± 1.58)
	2000	0.0	3.34 (± 0.08)	3.33 (± 0.08)	1.20 (± 0.46)	1.19 (± 0.45)	14.82 (± 1.71)	1.45 (± 0.96)
		0.3	3.36 (± 0.12)	3.25 (± 0.12)	1.39 (± 0.59)	1.35 (± 0.63)	14.94 (± 1.66)	4.05 (± 1.61)
		0.6	3.36 (± 0.12)	2.91 (± 0.11)	1.54 (± 0.64)	1.47 (± 0.70)	14.87 (± 1.93)	8.58 (± 1.36)
		0.9	3.36 (± 0.12)	2.12 (± 0.08)	1.69 (± 0.69)	1.58 (± 0.74)	14.79 (± 2.25)	13.19 (± 1.04)
Case 21	1000	0.0	6.60 (± 0.23)	6.71 (± 0.24)	2.21 (± 0.77)	2.23 (± 0.74)	19.12 (± 2.45)	2.91 (± 2.24)
		0.3	6.62 (± 0.19)	6.66 (± 0.17)	2.48 (± 1.09)	2.49 (± 1.09)	19.06 (± 3.18)	11.24 (± 3.14)
		0.6	6.63 (± 0.20)	6.39 (± 0.16)	2.72 (± 1.16)	2.69 (± 1.17)	19.13 (± 3.23)	22.90 (± 2.84)
		0.9	6.64 (± 0.22)	5.86 (± 0.16)	2.96 (± 1.20)	2.85 (± 1.23)	19.23 (± 3.26)	34.59 (± 2.55)
	2000	0.0	5.10 (± 0.11)	5.19 (± 0.12)	1.48 (± 0.55)	1.47 (± 0.57)	12.79 (± 1.33)	1.30 (± 0.97)
		0.3	5.11 (± 0.12)	5.14 (± 0.12)	1.40 (± 0.60)	1.37 (± 0.60)	13.23 (± 1.41)	8.47 (± 1.87)
		0.6	5.12 (± 0.13)	4.94 (± 0.11)	1.53 (± 0.66)	1.48 (± 0.64)	13.33 (± 1.53)	17.14 (± 1.83)
		0.9	5.12 (± 0.13)	4.53 (± 0.09)	1.65 (± 0.72)	1.58 (± 0.68)	13.45 (± 1.70)	25.86 (± 1.70)
Case 22	1000	0.0	4.97 (± 0.21)	4.98 (± 0.21)	2.08 (± 0.77)	2.10 (± 0.72)	33.11 (± 2.70)	2.93 (± 2.14)
		0.3	4.94 (± 0.19)	4.82 (± 0.16)	2.43 (± 1.03)	2.42 (± 1.04)	32.96 (± 3.54)	7.05 (± 3.01)
		0.6	4.95 (± 0.21)	4.37 (± 0.15)	2.67 (± 1.11)	2.61 (± 1.13)	32.92 (± 3.77)	14.53 (± 2.42)
		0.9	4.95 (± 0.25)	3.30 (± 0.16)	2.90 (± 1.16)	2.77 (± 1.13)	32.86 (± 4.03)	22.04 (± 1.74)
	2000	0.0	3.06 (± 0.09)	3.06 (± 0.09)	1.26 (± 0.44)	1.24 (± 0.45)	16.05 (± 1.84)	1.38 (± 0.82)
		0.3	3.08 (± 0.10)	2.96 (± 0.10)	1.35 (± 0.58)	1.32 (± 0.59)	16.08 (± 1.78)	3.71 (± 1.48)
		0.6	3.08 (± 0.09)	2.59 (± 0.08)	1.50 (± 0.64)	1.43 (± 0.65)	15.99 (± 2.05)	7.66 (± 1.36)
		0.9	3.09 (± 0.09)	1.66 (± 0.06)	1.64 (± 0.69)	1.53 (± 0.69)	15.89 (± 2.39)	11.73 (± 0.95)
Case 23	1000	0.0	7.26 (± 0.19)	7.59 (± 0.22)	2.34 (± 0.93)	2.14 (± 0.85)	25.26 (± 2.29)	3.63 (± 2.11)
		0.3	7.23 (± 0.23)	7.52 (± 0.24)	2.31 (± 0.75)	2.33 (± 0.83)	25.11 (± 2.60)	15.73 (± 3.80)
		0.6	7.23 (± 0.25)	7.36 (± 0.25)	2.51 (± 0.78)	2.51 (± 0.88)	25.06 (± 2.68)	32.31 (± 3.85)
		0.9	7.23 (± 0.26)	7.08 (± 0.26)	2.69 (± 0.80)	2.65 (± 0.96)	25.03 (± 2.79)	48.89 (± 3.85)
	2000	0.0	6.14 (± 0.13)	6.49 (± 0.14)	1.47 (± 0.49)	1.33 (± 0.44)	39.37 (± 2.68)	2.28 (± 1.74)
		0.3	6.16 (± 0.12)	6.46 (± 0.12)	1.52 (± 0.58)	1.39 (± 0.45)	38.85 (± 2.12)	14.64 (± 2.96)
		0.6	6.16 (± 0.12)	6.34 (± 0.12)	1.63 (± 0.62)	1.49 (± 0.48)	38.90 (± 2.11)	29.59 (± 2.81)
		0.9	6.15 (± 0.13)	6.12 (± 0.13)	1.74 (± 0.66)	1.58 (± 0.49)	39.00 (± 2.17)	44.58 (± 2.55)
Case 24	1000	0.0	5.57 (± 0.18)	5.63 (± 0.18)	1.88 (± 0.72)	1.85 (± 0.69)	12.29 (± 1.89)	2.83 (± 2.15)
		0.3	5.52 (± 0.18)	5.52 (± 0.18)	2.27 (± 0.86)	2.32 (± 0.90)	12.39 (± 2.44)	9.89 (± 3.12)
		0.6	5.51 (± 0.19)	5.28 (± 0.18)	2.45 (± 0.94)	2.47 (± 0.98)	12.43 (± 2.53)	20.64 (± 2.74)
		0.9	5.49 (± 0.21)	4.80 (± 0.18)	2.63 (± 0.98)	2.58 (± 1.03)	12.48 (± 2.66)	31.40 (± 2.19)
	2000	0.0	4.29 (± 0.09)	4.35 (± 0.10)	1.04 (± 0.40)	1.03 (± 0.38)	9.57 (± 1.10)	1.83 (± 1.18)
		0.3	4.30 (± 0.11)	4.32 (± 0.10)	1.39 (± 0.58)	1.36 (± 0.56)	9.76 (± 1.20)	7.66 (± 1.92)
		0.6	4.30 (± 0.11)	4.16 (± 0.09)	1.52 (± 0.63)	1.47 (± 0.61)	9.77 (± 1.36)	15.93 (± 1.74)
		0.9	4.29 (± 0.11)	3.83 (± 0.07)	1.65 (± 0.67)	1.58 (± 0.65)	9.79 (± 1.54)	24.27 (± 1.57)

TABLE 5
Estimation errors ($\times 10^{-2}$) for Cases 25–30.

Case	n	ρ	MIAE		MAE		MIAE	MAE
			$\hat{\mathbf{A}}^\dagger(t)$	$\hat{\mathbf{A}}^*(t)$	$\hat{\beta}^\dagger$	$\hat{\beta}^*$	$\hat{\sigma}^2(t)$	$\hat{\rho}$
Case 25	2000	0.0	3.51 (± 0.10)	3.53 (± 0.10)	1.55 (± 0.65)	1.53 (± 0.66)	18.86 (± 2.34)	1.96 (± 1.45)
		0.3	3.51 (± 0.10)	3.44 (± 0.10)	1.70 (± 0.71)	1.69 (± 0.71)	18.64 (± 2.18)	6.54 (± 2.61)
		0.6	3.51 (± 0.10)	3.17 (± 0.09)	1.86 (± 0.77)	1.83 (± 0.75)	18.61 (± 2.62)	13.45 (± 2.06)
		0.9	3.52 (± 0.11)	2.58 (± 0.08)	2.02 (± 0.83)	1.96 (± 0.77)	18.66 (± 2.97)	20.39 (± 1.34)
	5000	0.0	2.42 (± 0.06)	2.43 (± 0.06)	0.90 (± 0.36)	0.91 (± 0.37)	4.58 (± 0.92)	1.15 (± 0.78)
		0.3	2.39 (± 0.06)	2.35 (± 0.06)	0.95 (± 0.30)	0.94 (± 0.30)	4.61 (± 0.98)	3.99 (± 1.29)
		0.6	2.39 (± 0.06)	2.20 (± 0.06)	1.04 (± 0.34)	1.03 (± 0.32)	4.78 (± 1.16)	7.73 (± 1.04)
		0.9	2.39 (± 0.06)	1.89 (± 0.04)	1.14 (± 0.37)	1.11 (± 0.34)	5.03 (± 1.35)	11.42 (± 0.64)
Case 26	2000	0.0	3.46 (± 0.10)	3.48 (± 0.10)	1.52 (± 0.72)	1.50 (± 0.73)	19.09 (± 2.34)	1.68 (± 1.37)
		0.3	3.45 (± 0.09)	3.39 (± 0.10)	1.75 (± 0.68)	1.73 (± 0.67)	18.90 (± 2.18)	6.28 (± 2.39)
		0.6	3.45 (± 0.09)	3.11 (± 0.09)	1.91 (± 0.74)	1.87 (± 0.74)	18.89 (± 2.55)	13.04 (± 2.10)
		0.9	3.45 (± 0.10)	2.47 (± 0.08)	2.06 (± 0.80)	1.99 (± 0.78)	18.92 (± 2.91)	19.93 (± 1.51)
	5000	0.0	2.35 (± 0.05)	2.36 (± 0.05)	0.91 (± 0.38)	0.91 (± 0.39)	6.60 (± 0.93)	1.21 (± 0.73)
		0.3	2.33 (± 0.05)	2.29 (± 0.05)	0.92 (± 0.30)	0.91 (± 0.29)	6.61 (± 0.96)	3.91 (± 1.23)
		0.6	2.32 (± 0.05)	2.12 (± 0.04)	1.02 (± 0.32)	1.00 (± 0.31)	6.73 (± 1.09)	7.63 (± 0.99)
		0.9	2.32 (± 0.05)	1.74 (± 0.04)	1.12 (± 0.36)	1.08 (± 0.33)	6.88 (± 1.27)	11.32 (± 0.66)
Case 27	2000	0.0	4.56 (± 0.11)	4.60 (± 0.11)	1.55 (± 0.72)	1.55 (± 0.71)	22.37 (± 2.16)	1.89 (± 1.60)
		0.3	4.56 (± 0.11)	4.52 (± 0.10)	1.76 (± 0.79)	1.75 (± 0.82)	22.36 (± 2.18)	9.36 (± 2.40)
		0.6	4.62 (± 0.12)	4.25 (± 0.10)	1.93 (± 0.85)	1.89 (± 0.91)	31.24 (± 2.40)	18.48 (± 1.71)
		0.9	4.62 (± 0.13)	3.63 (± 0.10)	2.06 (± 0.90)	1.99 (± 0.94)	31.17 (± 2.80)	28.21 (± 1.02)
	5000	0.0	2.70 (± 0.05)	2.70 (± 0.05)	0.93 (± 0.31)	0.93 (± 0.33)	13.81 (± 1.23)	1.10 (± 0.68)
		0.3	2.68 (± 0.06)	2.61 (± 0.05)	0.95 (± 0.35)	0.94 (± 0.35)	13.72 (± 1.23)	4.90 (± 1.17)
		0.6	2.68 (± 0.06)	2.36 (± 0.05)	1.06 (± 0.37)	1.02 (± 0.37)	13.78 (± 1.33)	9.50 (± 0.95)
		0.9	2.78 (± 0.06)	1.68 (± 0.04)	1.16 (± 0.39)	1.10 (± 0.36)	19.45 (± 1.49)	14.72 (± 0.59)
Case 28	2000	0.0	3.13 (± 0.09)	3.14 (± 0.09)	1.51 (± 0.67)	1.49 (± 0.67)	22.56 (± 2.39)	1.83 (± 1.46)
		0.3	3.12 (± 0.08)	3.03 (± 0.08)	1.68 (± 0.70)	1.67 (± 0.71)	22.38 (± 2.22)	5.44 (± 2.37)
		0.6	3.13 (± 0.09)	2.71 (± 0.07)	1.85 (± 0.77)	1.81 (± 0.77)	22.33 (± 2.70)	11.14 (± 1.88)
		0.9	3.13 (± 0.10)	1.91 (± 0.06)	2.00 (± 0.82)	1.94 (± 0.80)	22.29 (± 3.24)	16.95 (± 1.02)
	5000	0.0	1.87 (± 0.04)	1.87 (± 0.04)	0.91 (± 0.33)	0.91 (± 0.34)	8.40 (± 1.33)	1.13 (± 0.74)
		0.3	1.84 (± 0.05)	1.78 (± 0.05)	0.93 (± 0.30)	0.92 (± 0.30)	8.34 (± 1.30)	2.81 (± 1.20)
		0.6	1.84 (± 0.06)	1.57 (± 0.05)	1.03 (± 0.33)	1.01 (± 0.32)	8.43 (± 1.43)	5.30 (± 0.92)
		0.9	1.84 (± 0.06)	1.06 (± 0.03)	1.13 (± 0.37)	1.10 (± 0.34)	8.52 (± 1.65)	7.72 (± 0.48)
Case 29	2000	0.0	5.38 (± 0.13)	5.41 (± 0.13)	1.69 (± 0.65)	1.69 (± 0.70)	22.85 (± 1.37)	2.04 (± 1.43)
		0.3	5.39 (± 0.15)	5.36 (± 0.15)	1.95 (± 0.86)	1.88 (± 0.93)	22.74 (± 1.56)	11.82 (± 2.49)
		0.6	5.39 (± 0.16)	5.12 (± 0.16)	2.09 (± 0.91)	2.01 (± 1.01)	22.77 (± 1.78)	24.36 (± 2.28)
		0.9	5.39 (± 0.18)	4.66 (± 0.17)	2.21 (± 0.95)	2.11 (± 1.05)	22.82 (± 2.03)	36.95 (± 2.04)
	5000	0.0	3.13 (± 0.05)	3.14 (± 0.05)	0.94 (± 0.42)	0.95 (± 0.43)	15.82 (± 1.17)	1.26 (± 0.66)
		0.3	3.11 (± 0.05)	3.04 (± 0.05)	1.00 (± 0.29)	0.99 (± 0.30)	15.76 (± 1.25)	6.00 (± 1.24)
		0.6	3.11 (± 0.06)	2.77 (± 0.04)	1.09 (± 0.32)	1.07 (± 0.31)	15.80 (± 1.34)	11.97 (± 1.03)
		0.9	3.10 (± 0.06)	2.16 (± 0.03)	1.19 (± 0.35)	1.14 (± 0.33)	15.84 (± 1.49)	17.89 (± 0.72)
Case 30	2000	0.0	4.42 (± 0.10)	4.44 (± 0.11)	1.60 (± 0.77)	1.62 (± 0.78)	24.10 (± 2.05)	1.86 (± 1.08)
		0.3	4.42 (± 0.11)	4.36 (± 0.12)	1.76 (± 0.73)	1.74 (± 0.74)	24.05 (± 2.19)	8.50 (± 2.63)
		0.6	4.42 (± 0.11)	4.10 (± 0.11)	1.92 (± 0.79)	1.88 (± 0.80)	23.99 (± 2.60)	17.91 (± 2.19)
		0.9	4.55 (± 0.13)	3.51 (± 0.11)	2.05 (± 0.87)	1.97 (± 0.88)	31.82 (± 2.90)	27.33 (± 1.36)
	5000	0.0	2.67 (± 0.05)	2.67 (± 0.05)	0.93 (± 0.35)	0.92 (± 0.35)	14.08 (± 1.23)	1.17 (± 0.77)
		0.3	2.65 (± 0.05)	2.58 (± 0.05)	0.97 (± 0.31)	0.95 (± 0.32)	14.02 (± 1.25)	4.64 (± 1.18)
		0.6	2.65 (± 0.05)	2.33 (± 0.05)	1.07 (± 0.34)	1.03 (± 0.33)	14.07 (± 1.36)	9.18 (± 0.97)
		0.9	2.78 (± 0.06)	1.67 (± 0.04)	1.17 (± 0.36)	1.10 (± 0.35)	19.51 (± 1.51)	14.59 (± 0.55)

TABLE 6
Estimation errors ($\times 10^{-2}$) for Cases 31–36.

Case	n	ρ	MIAE		MAE		MIAE	MAE
			$\hat{\mathbf{A}}^\dagger(t)$	$\hat{\mathbf{A}}^*(t)$	$\hat{\beta}^\dagger$	$\hat{\beta}^*$	$\hat{\sigma}^2(t)$	$\hat{\rho}$
Case 31	2000	0.0	3.52 ($\pm$ 0.10)	3.52 ($\pm$ 0.10)	1.54 ($\pm$ 0.66)	1.52 ($\pm$ 0.66)	22.39 ($\pm$ 2.22)	1.96 ($\pm$ 1.43)
		0.3	3.51 ($\pm$ 0.09)	3.43 ($\pm$ 0.09)	1.72 ($\pm$ 0.69)	1.63 ($\pm$ 0.70)	22.33 ($\pm$ 2.07)	6.34 ($\pm$ 2.58)
		0.6	3.51 ($\pm$ 0.10)	3.13 ($\pm$ 0.09)	1.88 ($\pm$ 0.74)	1.76 ($\pm$ 0.77)	22.27 ($\pm$ 2.51)	13.21 ($\pm$ 1.97)
		0.9	3.52 ($\pm$ 0.11)	2.46 ($\pm$ 0.08)	2.02 ($\pm$ 0.80)	1.89 ($\pm$ 0.78)	22.23 ($\pm$ 2.92)	20.14 ($\pm$ 1.26)
	5000	0.0	2.28 ($\pm$ 0.06)	2.28 ($\pm$ 0.06)	0.91 ($\pm$ 0.36)	0.90 ($\pm$ 0.37)	7.11 ($\pm$ 1.15)	1.16 ($\pm$ 0.78)
		0.3	2.24 ($\pm$ 0.06)	2.19 ($\pm$ 0.06)	0.93 ($\pm$ 0.30)	0.92 ($\pm$ 0.29)	7.10 ($\pm$ 1.21)	3.62 ($\pm$ 1.23)
		0.6	2.24 ($\pm$ 0.06)	2.02 ($\pm$ 0.05)	1.03 ($\pm$ 0.33)	1.01 ($\pm$ 0.31)	7.20 ($\pm$ 1.32)	7.03 ($\pm$ 0.99)
		0.9	2.24 ($\pm$ 0.06)	1.64 ($\pm$ 0.04)	1.12 ($\pm$ 0.36)	1.10 ($\pm$ 0.33)	7.36 ($\pm$ 1.50)	10.39 ($\pm$ 0.61)
Case 32	2000	0.0	3.46 ($\pm$ 0.10)	3.48 ($\pm$ 0.10)	1.50 ($\pm$ 0.70)	1.47 ($\pm$ 0.70)	22.57 ($\pm$ 2.21)	1.72 ($\pm$ 1.38)
		0.3	3.46 ($\pm$ 0.10)	3.39 ($\pm$ 0.10)	1.74 ($\pm$ 0.66)	1.64 ($\pm$ 0.69)	22.53 ($\pm$ 2.07)	6.13 ($\pm$ 2.33)
		0.6	3.46 ($\pm$ 0.10)	3.09 ($\pm$ 0.09)	1.91 ($\pm$ 0.71)	1.78 ($\pm$ 0.76)	22.45 ($\pm$ 2.52)	12.87 ($\pm$ 2.01)
		0.9	3.46 ($\pm$ 0.11)	2.39 ($\pm$ 0.08)	2.06 ($\pm$ 0.77)	1.90 ($\pm$ 0.80)	22.39 ($\pm$ 2.98)	19.77 ($\pm$ 1.44)
	5000	0.0	2.22 ($\pm$ 0.05)	2.23 ($\pm$ 0.05)	0.91 ($\pm$ 0.36)	0.90 ($\pm$ 0.38)	6.84 ($\pm$ 1.25)	1.25 ($\pm$ 0.73)
		0.3	2.20 ($\pm$ 0.05)	2.16 ($\pm$ 0.04)	0.91 ($\pm$ 0.31)	0.90 ($\pm$ 0.29)	6.79 ($\pm$ 1.18)	3.56 ($\pm$ 1.18)
		0.6	2.19 ($\pm$ 0.05)	1.97 ($\pm$ 0.04)	1.01 ($\pm$ 0.34)	0.99 ($\pm$ 0.31)	6.90 ($\pm$ 1.29)	6.97 ($\pm$ 0.94)
		0.9	2.19 ($\pm$ 0.05)	1.54 ($\pm$ 0.03)	1.11 ($\pm$ 0.36)	1.07 ($\pm$ 0.32)	7.05 ($\pm$ 1.46)	10.34 ($\pm$ 0.62)
Case 33	2000	0.0	4.56 ($\pm$ 0.11)	4.58 ($\pm$ 0.11)	1.55 ($\pm$ 0.70)	1.50 ($\pm$ 0.71)	22.24 ($\pm$ 2.16)	1.97 ($\pm$ 1.52)
		0.3	4.56 ($\pm$ 0.10)	4.51 ($\pm$ 0.10)	1.77 ($\pm$ 0.77)	1.72 ($\pm$ 0.82)	22.39 ($\pm$ 2.12)	9.33 ($\pm$ 2.34)
		0.6	4.61 ($\pm$ 0.12)	4.24 ($\pm$ 0.10)	1.94 ($\pm$ 0.83)	1.85 ($\pm$ 0.89)	31.19 ($\pm$ 2.41)	18.46 ($\pm$ 1.65)
		0.9	4.62 ($\pm$ 0.13)	3.62 ($\pm$ 0.09)	2.07 ($\pm$ 0.87)	1.94 ($\pm$ 0.92)	31.08 ($\pm$ 2.79)	28.22 ($\pm$ 1.05)
	5000	0.0	2.69 ($\pm$ 0.05)	2.69 ($\pm$ 0.05)	0.93 ($\pm$ 0.31)	0.92 ($\pm$ 0.34)	13.72 ($\pm$ 1.22)	1.15 ($\pm$ 0.76)
		0.3	2.68 ($\pm$ 0.06)	2.61 ($\pm$ 0.05)	0.95 ($\pm$ 0.35)	0.93 ($\pm$ 0.34)	13.66 ($\pm$ 1.25)	4.84 ($\pm$ 1.14)
		0.6	2.68 ($\pm$ 0.06)	2.36 ($\pm$ 0.05)	1.05 ($\pm$ 0.38)	1.01 ($\pm$ 0.35)	13.71 ($\pm$ 1.34)	9.45 ($\pm$ 0.94)
		0.9	2.77 ($\pm$ 0.06)	1.67 ($\pm$ 0.04)	1.16 ($\pm$ 0.39)	1.08 ($\pm$ 0.35)	19.35 ($\pm$ 1.49)	14.69 ($\pm$ 0.59)
Case 34	2000	0.0	3.25 ($\pm$ 0.09)	3.26 ($\pm$ 0.09)	1.49 ($\pm$ 0.68)	1.48 ($\pm$ 0.67)	24.79 ($\pm$ 2.24)	1.84 ($\pm$ 1.49)
		0.3	3.25 ($\pm$ 0.08)	3.15 ($\pm$ 0.08)	1.70 ($\pm$ 0.69)	1.61 ($\pm$ 0.72)	24.76 ($\pm$ 2.11)	5.60 ($\pm$ 2.34)
		0.6	3.25 ($\pm$ 0.09)	2.82 ($\pm$ 0.07)	1.86 ($\pm$ 0.75)	1.75 ($\pm$ 0.78)	24.68 ($\pm$ 2.57)	11.64 ($\pm$ 1.82)
		0.9	3.26 ($\pm$ 0.10)	2.00 ($\pm$ 0.06)	2.01 ($\pm$ 0.80)	1.88 ($\pm$ 0.81)	24.61 ($\pm$ 3.06)	17.83 ($\pm$ 1.03)
	5000	0.0	1.89 ($\pm$ 0.05)	1.89 ($\pm$ 0.04)	0.91 ($\pm$ 0.33)	0.90 ($\pm$ 0.34)	9.48 ($\pm$ 1.31)	1.20 ($\pm$ 0.72)
		0.3	1.86 ($\pm$ 0.05)	1.80 ($\pm$ 0.05)	0.92 ($\pm$ 0.31)	0.91 ($\pm$ 0.29)	9.44 ($\pm$ 1.26)	2.82 ($\pm$ 1.16)
		0.6	1.86 ($\pm$ 0.06)	1.57 ($\pm$ 0.05)	1.02 ($\pm$ 0.34)	1.00 ($\pm$ 0.31)	9.53 ($\pm$ 1.37)	5.38 ($\pm$ 0.90)
		0.9	1.86 ($\pm$ 0.06)	1.02 ($\pm$ 0.03)	1.11 ($\pm$ 0.37)	1.08 ($\pm$ 0.32)	9.64 ($\pm$ 1.55)	7.89 ($\pm$ 0.50)
Case 35	2000	0.0	5.34 ($\pm$ 0.12)	5.38 ($\pm$ 0.13)	1.69 ($\pm$ 0.63)	1.64 ($\pm$ 0.68)	20.41 ($\pm$ 1.81)	1.99 ($\pm$ 1.48)
		0.3	5.36 ($\pm$ 0.15)	5.32 ($\pm$ 0.15)	1.96 ($\pm$ 0.82)	1.81 ($\pm$ 0.90)	20.48 ($\pm$ 2.03)	11.74 ($\pm$ 2.52)
		0.6	5.36 ($\pm$ 0.16)	5.09 ($\pm$ 0.16)	2.10 ($\pm$ 0.87)	1.94 ($\pm$ 0.98)	20.44 ($\pm$ 2.30)	24.30 ($\pm$ 2.31)
		0.9	5.36 ($\pm$ 0.18)	4.63 ($\pm$ 0.16)	2.22 ($\pm$ 0.91)	2.03 ($\pm$ 1.03)	20.40 ($\pm$ 2.57)	36.92 ($\pm$ 2.09)
	5000	0.0	3.11 ($\pm$ 0.05)	3.11 ($\pm$ 0.05)	0.94 ($\pm$ 0.40)	0.93 ($\pm$ 0.43)	15.72 ($\pm$ 1.16)	1.31 ($\pm$ 0.71)
		0.3	3.09 ($\pm$ 0.05)	3.02 ($\pm$ 0.05)	0.98 ($\pm$ 0.30)	0.97 ($\pm$ 0.31)	15.70 ($\pm$ 1.29)	5.98 ($\pm$ 1.21)
		0.6	3.08 ($\pm$ 0.05)	2.75 ($\pm$ 0.04)	1.08 ($\pm$ 0.33)	1.05 ($\pm$ 0.32)	15.72 ($\pm$ 1.37)	11.95 ($\pm$ 1.01)
		0.9	3.08 ($\pm$ 0.06)	2.15 ($\pm$ 0.03)	1.17 ($\pm$ 0.36)	1.12 ($\pm$ 0.33)	15.75 ($\pm$ 1.50)	17.89 ($\pm$ 0.70)
Case 36	2000	0.0	4.40 ($\pm$ 0.10)	4.42 ($\pm$ 0.10)	1.61 ($\pm$ 0.74)	1.61 ($\pm$ 0.76)	23.98 ($\pm$ 2.01)	1.83 ($\pm$ 1.08)
		0.3	4.40 ($\pm$ 0.11)	4.35 ($\pm$ 0.12)	1.79 ($\pm$ 0.70)	1.68 ($\pm$ 0.72)	24.06 ($\pm$ 2.15)	8.40 ($\pm$ 2.57)
		0.6	4.40 ($\pm$ 0.11)	4.08 ($\pm$ 0.11)	1.94 ($\pm$ 0.76)	1.81 ($\pm$ 0.79)	23.96 ($\pm$ 2.56)	17.82 ($\pm$ 2.12)
		0.9	4.53 ($\pm$ 0.13)	3.50 ($\pm$ 0.11)	2.08 ($\pm$ 0.84)	1.90 ($\pm$ 0.87)	31.74 ($\pm$ 2.88)	27.29 ($\pm$ 1.34)
	5000	0.0	2.66 ($\pm$ 0.05)	2.66 ($\pm$ 0.05)	0.93 ($\pm$ 0.34)	0.91 ($\pm$ 0.36)	13.99 ($\pm$ 1.23)	1.23 ($\pm$ 0.78)
		0.3	2.64 ($\pm$ 0.05)	2.57 ($\pm$ 0.05)	0.96 ($\pm$ 0.31)	0.94 ($\pm$ 0.31)	13.96 ($\pm$ 1.28)	4.60 ($\pm$ 1.14)
		0.6	2.64 ($\pm$ 0.05)	2.32 ($\pm$ 0.05)	1.06 ($\pm$ 0.34)	1.02 ($\pm$ 0.32)	14.00 ($\pm$ 1.38)	9.15 ($\pm$ 0.92)
		0.9	2.76 ($\pm$ 0.06)	1.66 ($\pm$ 0.04)	1.16 ($\pm$ 0.37)	1.09 ($\pm$ 0.34)	19.41 ($\pm$ 1.51)	14.58 ($\pm$ 0.51)

TABLE 7
Estimation errors ($\times 10^{-2}$) for Cases 37–42.

Case	n	ρ	MIAE		MAE		MIAE	MAE
			$\hat{\mathbf{A}}^\dagger(t)$	$\hat{\mathbf{A}}^*(t)$	$\hat{\beta}^\dagger$	$\hat{\beta}^*$	$\hat{\sigma}^2(t)$	$\hat{\rho}$
Case 37	2000	0.0	3.60 (± 0.09)	3.60 (± 0.09)	1.51 (± 0.68)	1.52 (± 0.68)	23.71 (± 2.11)	1.91 (± 1.38)
		0.3	3.59 (± 0.10)	3.50 (± 0.10)	1.70 (± 0.71)	1.67 (± 0.70)	23.57 (± 1.96)	6.71 (± 2.51)
		0.6	3.59 (± 0.10)	3.20 (± 0.09)	1.87 (± 0.77)	1.81 (± 0.76)	23.57 (± 2.36)	13.85 (± 1.93)
		0.9	3.59 (± 0.11)	2.53 (± 0.08)	2.02 (± 0.82)	1.93 (± 0.78)	23.61 (± 2.78)	21.00 (± 1.23)
	5000	0.0	2.32 (± 0.06)	2.31 (± 0.06)	0.92 (± 0.37)	0.89 (± 0.36)	8.74 (± 1.05)	1.06 (± 0.74)
		0.3	2.29 (± 0.06)	2.23 (± 0.06)	0.94 (± 0.30)	0.93 (± 0.29)	8.81 (± 1.05)	3.82 (± 1.32)
		0.6	2.28 (± 0.06)	2.05 (± 0.05)	1.04 (± 0.33)	1.02 (± 0.31)	8.95 (± 1.24)	7.32 (± 1.06)
		0.9	2.28 (± 0.06)	1.66 (± 0.04)	1.14 (± 0.37)	1.10 (± 0.33)	9.11 (± 1.49)	10.76 (± 0.63)
Case 38	2000	0.0	3.56 (± 0.10)	3.57 (± 0.10)	1.48 (± 0.73)	1.49 (± 0.72)	23.95 (± 2.07)	1.66 (± 1.32)
		0.3	3.56 (± 0.10)	3.47 (± 0.10)	1.73 (± 0.69)	1.70 (± 0.69)	23.82 (± 1.90)	6.48 (± 2.41)
		0.6	3.56 (± 0.10)	3.17 (± 0.09)	1.89 (± 0.75)	1.83 (± 0.75)	23.85 (± 2.25)	13.54 (± 1.96)
		0.9	3.56 (± 0.11)	2.46 (± 0.08)	2.05 (± 0.80)	1.95 (± 0.80)	23.91 (± 2.65)	20.66 (± 1.36)
	5000	0.0	2.28 (± 0.06)	2.27 (± 0.05)	0.92 (± 0.38)	0.90 (± 0.37)	10.41 (± 1.04)	1.15 (± 0.71)
		0.3	2.26 (± 0.05)	2.20 (± 0.04)	0.92 (± 0.31)	0.92 (± 0.28)	10.45 (± 1.07)	3.76 (± 1.24)
		0.6	2.26 (± 0.05)	2.00 (± 0.04)	1.02 (± 0.34)	1.00 (± 0.31)	10.56 (± 1.24)	7.25 (± 0.99)
		0.9	2.25 (± 0.05)	1.55 (± 0.03)	1.12 (± 0.36)	1.08 (± 0.32)	10.69 (± 1.48)	10.69 (± 0.62)
Case 39	2000	0.0	4.63 (± 0.11)	4.66 (± 0.11)	1.54 (± 0.71)	1.52 (± 0.71)	23.47 (± 2.06)	1.91 (± 1.56)
		0.3	4.63 (± 0.11)	4.59 (± 0.10)	1.75 (± 0.81)	1.73 (± 0.80)	23.53 (± 2.03)	9.63 (± 2.37)
		0.6	4.69 (± 0.13)	4.32 (± 0.11)	1.93 (± 0.87)	1.86 (± 0.88)	32.34 (± 2.26)	18.91 (± 1.69)
		0.9	4.69 (± 0.14)	3.70 (± 0.10)	2.07 (± 0.91)	1.95 (± 0.91)	32.29 (± 2.68)	28.85 (± 1.03)
	5000	0.0	2.72 (± 0.05)	2.72 (± 0.05)	0.94 (± 0.33)	0.92 (± 0.31)	14.38 (± 1.22)	1.07 (± 0.70)
		0.3	2.70 (± 0.06)	2.63 (± 0.05)	0.95 (± 0.36)	0.94 (± 0.34)	14.32 (± 1.21)	5.05 (± 1.21)
		0.6	2.70 (± 0.06)	2.38 (± 0.05)	1.05 (± 0.38)	1.02 (± 0.36)	14.40 (± 1.34)	9.74 (± 0.98)
		0.9	2.80 (± 0.06)	1.69 (± 0.04)	1.16 (± 0.40)	1.09 (± 0.35)	20.05 (± 1.52)	14.99 (± 0.61)
Case 40	2000	0.0	3.33 (± 0.09)	3.33 (± 0.09)	1.48 (± 0.70)	1.49 (± 0.69)	26.11 (± 2.14)	1.81 (± 1.44)
		0.3	3.32 (± 0.09)	3.23 (± 0.08)	1.69 (± 0.71)	1.65 (± 0.71)	25.99 (± 2.02)	5.93 (± 2.42)
		0.6	3.32 (± 0.09)	2.89 (± 0.08)	1.86 (± 0.77)	1.79 (± 0.78)	25.97 (± 2.46)	12.28 (± 1.82)
		0.9	3.33 (± 0.11)	2.06 (± 0.07)	2.01 (± 0.82)	1.91 (± 0.80)	25.98 (± 2.93)	18.68 (± 1.01)
	5000	0.0	1.93 (± 0.04)	1.93 (± 0.04)	0.92 (± 0.34)	0.89 (± 0.33)	10.46 (± 1.17)	1.09 (± 0.70)
		0.3	1.90 (± 0.05)	1.83 (± 0.05)	0.93 (± 0.31)	0.92 (± 0.29)	10.47 (± 1.14)	3.03 (± 1.25)
		0.6	1.90 (± 0.06)	1.60 (± 0.05)	1.03 (± 0.34)	1.01 (± 0.31)	10.59 (± 1.30)	5.68 (± 0.96)
		0.9	1.90 (± 0.06)	1.04 (± 0.03)	1.13 (± 0.37)	1.09 (± 0.33)	10.72 (± 1.54)	8.25 (± 0.51)
Case 41	2000	0.0	5.45 (± 0.13)	5.49 (± 0.13)	1.68 (± 0.66)	1.67 (± 0.70)	26.59 (± 1.30)	2.02 (± 1.32)
		0.3	5.47 (± 0.16)	5.44 (± 0.16)	1.95 (± 0.88)	1.86 (± 0.92)	26.57 (± 1.45)	12.05 (± 2.44)
		0.6	5.47 (± 0.17)	5.20 (± 0.16)	2.09 (± 0.94)	1.98 (± 1.00)	26.61 (± 1.65)	24.82 (± 2.24)
		0.9	5.48 (± 0.18)	4.74 (± 0.17)	2.22 (± 0.97)	2.09 (± 1.04)	26.69 (± 1.91)	37.63 (± 2.03)
	5000	0.0	3.16 (± 0.05)	3.16 (± 0.05)	0.95 (± 0.43)	0.94 (± 0.41)	16.81 (± 1.06)	1.23 (± 0.66)
		0.3	3.14 (± 0.05)	3.06 (± 0.05)	0.99 (± 0.31)	0.98 (± 0.28)	16.77 (± 1.09)	6.14 (± 1.25)
		0.6	3.13 (± 0.06)	2.80 (± 0.04)	1.09 (± 0.34)	1.06 (± 0.30)	16.83 (± 1.21)	12.18 (± 1.05)
		0.9	3.13 (± 0.06)	2.18 (± 0.03)	1.18 (± 0.36)	1.13 (± 0.32)	16.90 (± 1.38)	18.17 (± 0.74)
Case 42	2000	0.0	4.46 (± 0.10)	4.48 (± 0.11)	1.58 (± 0.78)	1.59 (± 0.76)	25.20 (± 1.96)	1.81 (± 1.11)
		0.3	4.46 (± 0.11)	4.41 (± 0.12)	1.76 (± 0.73)	1.72 (± 0.73)	25.20 (± 2.05)	8.76 (± 2.61)
		0.6	4.46 (± 0.11)	4.14 (± 0.11)	1.92 (± 0.79)	1.85 (± 0.80)	25.16 (± 2.46)	18.40 (± 2.15)
		0.9	4.60 (± 0.13)	3.57 (± 0.12)	2.05 (± 0.88)	1.94 (± 0.87)	32.94 (± 2.77)	27.95 (± 1.33)
	5000	0.0	2.68 (± 0.05)	2.68 (± 0.05)	0.93 (± 0.36)	0.90 (± 0.34)	14.65 (± 1.22)	1.15 (± 0.76)
		0.3	2.66 (± 0.05)	2.59 (± 0.05)	0.96 (± 0.32)	0.94 (± 0.31)	14.62 (± 1.25)	4.79 (± 1.21)
		0.6	2.65 (± 0.05)	2.33 (± 0.05)	1.06 (± 0.35)	1.02 (± 0.32)	14.69 (± 1.37)	9.41 (± 0.98)
		0.9	2.79 (± 0.06)	1.67 (± 0.04)	1.17 (± 0.38)	1.09 (± 0.34)	20.11 (± 1.53)	14.86 (± 0.55)

TABLE 8
Estimation errors ($\times 10^{-2}$) for Cases 43–48.

Case	n	ρ	MIAE		MAE		MIAE	MAE
			$\hat{\mathbf{A}}^\dagger(t)$	$\hat{\mathbf{A}}^*(t)$	$\hat{\beta}^\dagger$	$\hat{\beta}^*$	$\hat{\sigma}^2(t)$	$\hat{\rho}$
Case 43	2000	0.0	3.03 ($\pm$ 0.09)	3.04 ($\pm$ 0.09)	1.38 ($\pm$ 0.59)	1.29 ($\pm$ 0.55)	15.01 ($\pm$ 1.82)	2.13 ($\pm$ 1.48)
		0.3	3.02 ($\pm$ 0.08)	2.96 ($\pm$ 0.08)	1.48 ($\pm$ 0.61)	1.41 ($\pm$ 0.60)	14.82 ($\pm$ 1.76)	6.41 ($\pm$ 2.66)
		0.6	3.03 ($\pm$ 0.09)	2.72 ($\pm$ 0.08)	1.61 ($\pm$ 0.68)	1.53 ($\pm$ 0.64)	14.76 ($\pm$ 2.15)	13.17 ($\pm$ 2.18)
		0.9	3.03 ($\pm$ 0.09)	2.19 ($\pm$ 0.07)	1.74 ($\pm$ 0.74)	1.64 ($\pm$ 0.65)	14.73 ($\pm$ 2.57)	20.00 ($\pm$ 1.39)
	5000	0.0	2.03 ($\pm$ 0.05)	2.04 ($\pm$ 0.05)	0.78 ($\pm$ 0.30)	0.77 ($\pm$ 0.32)	3.64 ($\pm$ 0.97)	1.21 ($\pm$ 0.88)
		0.3	2.00 ($\pm$ 0.05)	1.98 ($\pm$ 0.05)	0.82 ($\pm$ 0.26)	0.79 ($\pm$ 0.25)	3.57 ($\pm$ 0.92)	3.76 ($\pm$ 1.33)
		0.6	2.00 ($\pm$ 0.05)	1.84 ($\pm$ 0.05)	0.91 ($\pm$ 0.29)	0.86 ($\pm$ 0.27)	3.67 ($\pm$ 0.98)	7.37 ($\pm$ 1.07)
		0.9	2.00 ($\pm$ 0.05)	1.57 ($\pm$ 0.03)	0.99 ($\pm$ 0.31)	0.93 ($\pm$ 0.29)	3.81 ($\pm$ 1.07)	10.93 ($\pm$ 0.66)
Case 44	2000	0.0	2.98 ($\pm$ 0.09)	3.00 ($\pm$ 0.09)	1.33 ($\pm$ 0.62)	1.24 ($\pm$ 0.61)	15.17 ($\pm$ 1.84)	1.86 ($\pm$ 1.54)
		0.3	2.98 ($\pm$ 0.08)	2.92 ($\pm$ 0.09)	1.50 ($\pm$ 0.59)	1.43 ($\pm$ 0.59)	15.01 ($\pm$ 1.79)	6.22 ($\pm$ 2.33)
		0.6	2.98 ($\pm$ 0.09)	2.68 ($\pm$ 0.08)	1.64 ($\pm$ 0.64)	1.54 ($\pm$ 0.65)	14.96 ($\pm$ 2.16)	12.81 ($\pm$ 2.16)
		0.9	2.98 ($\pm$ 0.09)	2.11 ($\pm$ 0.07)	1.78 ($\pm$ 0.70)	1.65 ($\pm$ 0.66)	14.92 ($\pm$ 2.56)	19.60 ($\pm$ 1.53)
	5000	0.0	1.98 ($\pm$ 0.04)	1.99 ($\pm$ 0.04)	0.78 ($\pm$ 0.31)	0.77 ($\pm$ 0.33)	4.30 ($\pm$ 0.79)	1.32 ($\pm$ 0.69)
		0.3	1.96 ($\pm$ 0.04)	1.93 ($\pm$ 0.04)	0.81 ($\pm$ 0.25)	0.76 ($\pm$ 0.26)	4.23 ($\pm$ 0.81)	3.70 ($\pm$ 1.30)
		0.6	1.95 ($\pm$ 0.04)	1.78 ($\pm$ 0.04)	0.90 ($\pm$ 0.27)	0.83 ($\pm$ 0.28)	4.30 ($\pm$ 0.88)	7.30 ($\pm$ 1.05)
		0.9	1.95 ($\pm$ 0.05)	1.44 ($\pm$ 0.03)	0.98 ($\pm$ 0.30)	0.89 ($\pm$ 0.29)	4.41 ($\pm$ 0.98)	10.86 ($\pm$ 0.69)
Case 45	2000	0.0	4.00 ($\pm$ 0.10)	4.01 ($\pm$ 0.10)	1.37 ($\pm$ 0.61)	1.34 ($\pm$ 0.58)	21.36 ($\pm$ 1.46)	2.23 ($\pm$ 1.52)
		0.3	4.00 ($\pm$ 0.09)	3.94 ($\pm$ 0.09)	1.55 ($\pm$ 0.65)	1.50 ($\pm$ 0.71)	21.30 ($\pm$ 1.65)	9.07 ($\pm$ 2.43)
		0.6	4.00 ($\pm$ 0.10)	3.70 ($\pm$ 0.08)	1.68 ($\pm$ 0.71)	1.60 ($\pm$ 0.78)	21.23 ($\pm$ 1.96)	18.91 ($\pm$ 1.81)
		0.9	4.00 ($\pm$ 0.11)	3.21 ($\pm$ 0.08)	1.79 ($\pm$ 0.77)	1.67 ($\pm$ 0.80)	21.17 ($\pm$ 2.27)	28.79 ($\pm$ 1.11)
	5000	0.0	2.38 ($\pm$ 0.05)	2.38 ($\pm$ 0.05)	0.80 ($\pm$ 0.25)	0.79 ($\pm$ 0.29)	9.03 ($\pm$ 0.96)	1.14 ($\pm$ 0.74)
		0.3	2.37 ($\pm$ 0.05)	2.31 ($\pm$ 0.05)	0.83 ($\pm$ 0.29)	0.79 ($\pm$ 0.31)	8.95 ($\pm$ 0.96)	5.00 ($\pm$ 1.20)
		0.6	2.40 ($\pm$ 0.05)	2.08 ($\pm$ 0.04)	0.92 ($\pm$ 0.31)	0.86 ($\pm$ 0.32)	13.69 ($\pm$ 1.03)	9.88 ($\pm$ 0.93)
		0.9	2.40 ($\pm$ 0.05)	1.48 ($\pm$ 0.03)	1.00 ($\pm$ 0.32)	0.91 ($\pm$ 0.31)	13.71 ($\pm$ 1.14)	14.73 ($\pm$ 0.59)
Case 46	2000	0.0	2.75 ($\pm$ 0.08)	2.75 ($\pm$ 0.08)	1.34 ($\pm$ 0.60)	1.24 ($\pm$ 0.56)	17.41 ($\pm$ 1.85)	1.98 ($\pm$ 1.59)
		0.3	2.74 ($\pm$ 0.07)	2.66 ($\pm$ 0.07)	1.46 ($\pm$ 0.61)	1.40 ($\pm$ 0.62)	17.24 ($\pm$ 1.81)	5.47 ($\pm$ 2.38)
		0.6	2.75 ($\pm$ 0.08)	2.38 ($\pm$ 0.06)	1.60 ($\pm$ 0.68)	1.52 ($\pm$ 0.66)	17.19 ($\pm$ 2.18)	11.14 ($\pm$ 2.00)
		0.9	2.75 ($\pm$ 0.09)	1.69 ($\pm$ 0.05)	1.74 ($\pm$ 0.73)	1.62 ($\pm$ 0.67)	17.15 ($\pm$ 2.60)	16.99 ($\pm$ 1.07)
	5000	0.0	1.62 ($\pm$ 0.04)	1.62 ($\pm$ 0.04)	0.78 ($\pm$ 0.28)	0.77 ($\pm$ 0.30)	6.40 ($\pm$ 1.03)	1.20 ($\pm$ 0.78)
		0.3	1.60 ($\pm$ 0.05)	1.55 ($\pm$ 0.04)	0.81 ($\pm$ 0.25)	0.77 ($\pm$ 0.26)	6.33 ($\pm$ 0.98)	2.74 ($\pm$ 1.19)
		0.6	1.60 ($\pm$ 0.05)	1.36 ($\pm$ 0.04)	0.90 ($\pm$ 0.28)	0.85 ($\pm$ 0.28)	6.39 ($\pm$ 1.06)	5.22 ($\pm$ 0.96)
		0.9	1.60 ($\pm$ 0.05)	0.91 ($\pm$ 0.03)	0.98 ($\pm$ 0.31)	0.92 ($\pm$ 0.29)	6.46 ($\pm$ 1.21)	7.66 ($\pm$ 0.51)
Case 47	2000	0.0	4.80 ($\pm$ 0.11)	4.84 ($\pm$ 0.11)	1.49 ($\pm$ 0.58)	1.44 ($\pm$ 0.58)	15.36 ($\pm$ 1.29)	2.16 ($\pm$ 1.55)
		0.3	4.81 ($\pm$ 0.13)	4.79 ($\pm$ 0.13)	1.71 ($\pm$ 0.73)	1.59 ($\pm$ 0.81)	15.14 ($\pm$ 1.36)	12.72 ($\pm$ 2.63)
		0.6	4.81 ($\pm$ 0.14)	4.61 ($\pm$ 0.14)	1.83 ($\pm$ 0.77)	1.71 ($\pm$ 0.87)	15.17 ($\pm$ 1.47)	26.05 ($\pm$ 2.49)
		0.9	4.81 ($\pm$ 0.15)	4.26 ($\pm$ 0.15)	1.93 ($\pm$ 0.81)	1.80 ($\pm$ 0.91)	15.22 ($\pm$ 1.61)	39.43 ($\pm$ 2.29)
	5000	0.0	2.77 ($\pm$ 0.05)	2.77 ($\pm$ 0.05)	0.82 ($\pm$ 0.33)	0.80 ($\pm$ 0.38)	10.16 ($\pm$ 0.87)	1.34 ($\pm$ 0.74)
		0.3	2.75 ($\pm$ 0.05)	2.69 ($\pm$ 0.04)	0.87 ($\pm$ 0.24)	0.83 ($\pm$ 0.26)	10.10 ($\pm$ 0.96)	6.26 ($\pm$ 1.32)
		0.6	2.74 ($\pm$ 0.05)	2.48 ($\pm$ 0.04)	0.96 ($\pm$ 0.27)	0.90 ($\pm$ 0.27)	10.12 ($\pm$ 1.02)	12.57 ($\pm$ 1.10)
		0.9	2.74 ($\pm$ 0.05)	1.99 ($\pm$ 0.03)	1.04 ($\pm$ 0.29)	0.96 ($\pm$ 0.28)	10.15 ($\pm$ 1.13)	18.86 ($\pm$ 0.77)
Case 48	2000	0.0	3.93 ($\pm$ 0.09)	3.94 ($\pm$ 0.09)	1.41 ($\pm$ 0.67)	1.40 ($\pm$ 0.68)	15.22 ($\pm$ 1.60)	1.94 ($\pm$ 1.20)
		0.3	3.93 ($\pm$ 0.10)	3.88 ($\pm$ 0.10)	1.54 ($\pm$ 0.63)	1.48 ($\pm$ 0.61)	15.11 ($\pm$ 1.83)	8.87 ($\pm$ 2.71)
		0.6	3.96 ($\pm$ 0.10)	3.65 ($\pm$ 0.10)	1.65 ($\pm$ 0.72)	1.57 ($\pm$ 0.70)	21.89 ($\pm$ 2.07)	18.05 ($\pm$ 2.10)
		0.9	3.95 ($\pm$ 0.11)	3.11 ($\pm$ 0.09)	1.77 ($\pm$ 0.77)	1.65 ($\pm$ 0.73)	21.83 ($\pm$ 2.36)	27.72 ($\pm$ 1.50)
	5000	0.0	2.37 ($\pm$ 0.04)	2.36 ($\pm$ 0.04)	0.79 ($\pm$ 0.29)	0.79 ($\pm$ 0.30)	9.31 ($\pm$ 0.96)	1.27 ($\pm$ 0.78)
		0.3	2.35 ($\pm$ 0.05)	2.29 ($\pm$ 0.04)	0.84 ($\pm$ 0.26)	0.80 ($\pm$ 0.28)	9.26 ($\pm$ 0.96)	4.70 ($\pm$ 1.24)
		0.6	2.35 ($\pm$ 0.05)	2.09 ($\pm$ 0.04)	0.93 ($\pm$ 0.28)	0.86 ($\pm$ 0.29)	9.29 ($\pm$ 1.02)	9.40 ($\pm$ 1.02)
		0.9	2.40 ($\pm$ 0.05)	1.49 ($\pm$ 0.03)	1.01 ($\pm$ 0.30)	0.92 ($\pm$ 0.30)	13.77 ($\pm$ 1.15)	14.57 ($\pm$ 0.58)

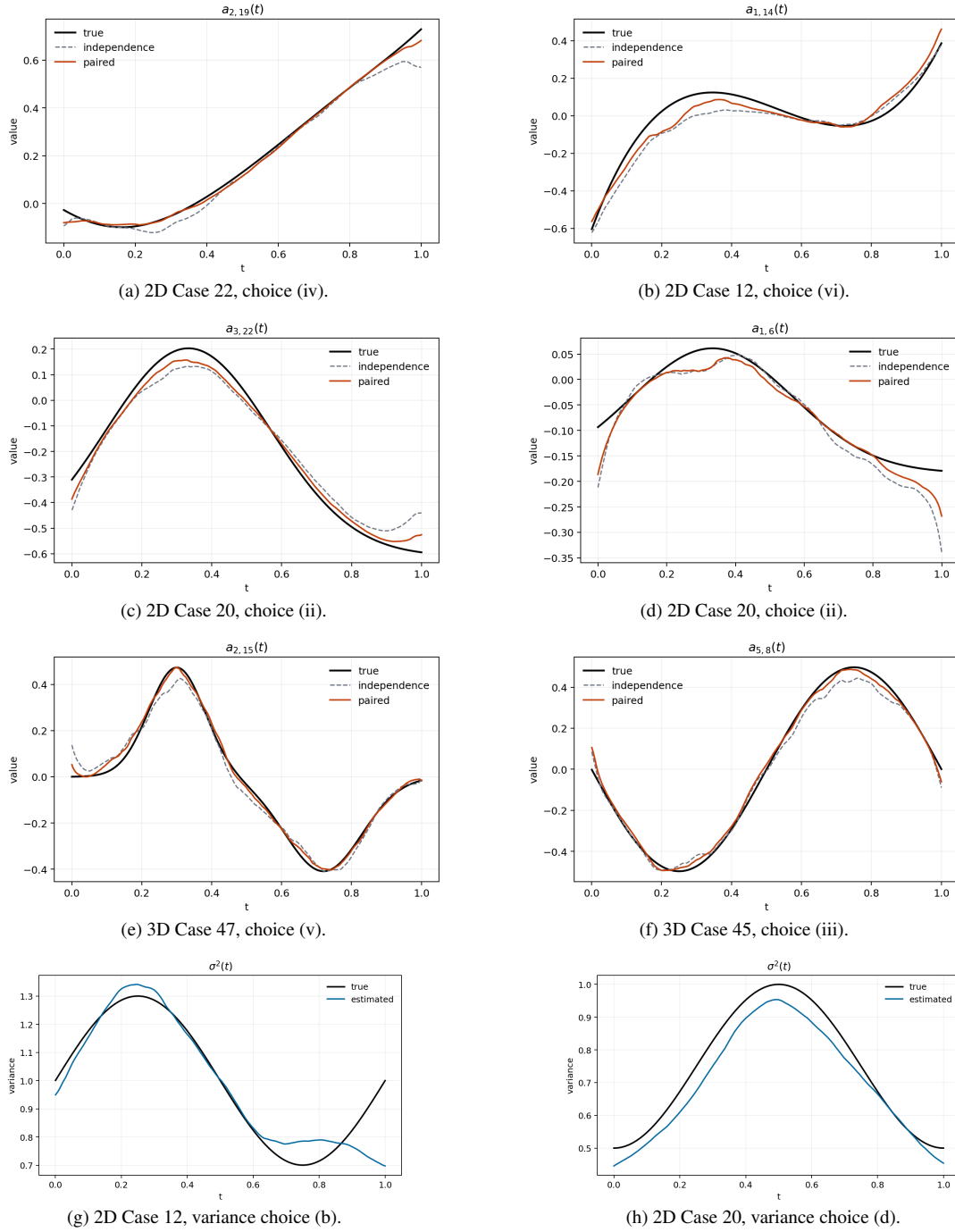


Fig 2: Representative fitted-versus-true functions: panels (a)–(f) show coefficient functions, and panels (g)–(h) show variance functions.

5. Glaucoma management with fundus images. Our analysis uses 273 matched OD–OS visit pairs from 119 subjects in the GRAPE study to examine how the association between fundus images and IOP varies with age after adjusting for gender and 59 visual-field measurements. Each pair contains OD and OS records from the same subject at the same follow-up time; let $i = 1, \dots, 273$ index paired visits and $j \in \{1, 2\}$ index the two eyes. Figure 1(a)–(b) shows representative CFP and optic-disc-centered ROI images. The images are resized to $\mathcal{X}_{ij} \in \mathbb{R}^{192 \times 192 \times 3}$, and OS images are horizontally flipped to align with OD while retaining their paired identity. We analyze CFP and ROI as separate image representations, with particular interest in the ROI because major fundus manifestations of glaucoma concentrate around the optic disc (Prum et al., 2016). For each representation, blockwise rank- R CP decomposition yields $\mathbf{X}_{ij}^* \in \mathbb{R}^{R \times S}$, as described in Section 2.2.

Our hypothesis is that the association between IOP and fundus images may vary with different ages, thus we consider model (3) with age being the index variable t_i . The one-way covariate $\mathbf{z}_i$ includes subject’s gender and 59 view field values (cf. locations in Figure 1(c)). To achieve consistency under the same location for OD and OS image, we record VF value along trajectory of scanning lines, as shown in Figure 1(c). All continuous variables are centered and standardized. In model (3), j indexes the paired eyes, whose errors follow (4) with variance $\sigma^2(t_i)$ and within-pair correlation ρ . Separate five-fold cross-validation for CFP and ROI selects the partition grid, CP rank R , coefficient bandwidth h , and variance bandwidth $\bar{h}$. Across 56 available (S, R) configurations, we examine 32 values of h from 0.18 to 8.00 and 15 values of $\bar{h}$ from 0.20 to 4.00. For each (S, R) , Table 9 reports the smallest five-fold cross-validation RMSE among the 480 candidate $(h, \bar{h})$ combinations.

TABLE 9
Five-fold cross-validation RMSE by tensor partition and CP rank.

$S^{(1)} \times S^{(2)} \times S^{(3)}$	CFP				ROI			
	$R=1$	$R=2$	$R=3$	$R=4$	$R=1$	$R=2$	$R=3$	$R=4$
$2 \times 2 \times 1$	0.9958	0.9993	1.0068	1.0125	0.9793	0.9746	0.9942	0.9964
$2 \times 3 \times 1$	1.0032	1.0213	1.0207	–	0.9849	0.9942	1.0219	–
$2 \times 4 \times 1$	1.0096	1.0219	1.0212	–	0.9924	1.0089	1.0370	–
$2 \times 6 \times 1$	1.0204	1.0369	1.1059	–	1.0093	1.0410	1.0448	–
$3 \times 2 \times 1$	0.9931	0.9993	1.0247	–	0.9742	0.9936	1.0190	–
$3 \times 3 \times 1$	0.9964	1.0261	1.0383	1.0440	0.9782	0.9978	1.0563	1.0698
$3 \times 4 \times 1$	0.9890	1.0025	1.0547	–	1.0054	1.0521	1.0905	–
$3 \times 6 \times 1$	1.0087	1.0596	1.1367	–	1.0112	1.0890	1.1446	–
$4 \times 2 \times 1$	1.0055	1.0135	1.0326	–	0.9766	0.9885	1.0242	–
$4 \times 3 \times 1$	1.0054	1.0626	1.1094	–	0.9823	1.0246	1.0476	–
$4 \times 4 \times 1$	1.0131	1.0382	1.1292	1.1915	1.0023	1.0397	1.1207	1.1167
$4 \times 6 \times 1$	1.0363	1.1582	1.2898	–	0.9768	1.0598	1.1639	–
$6 \times 2 \times 1$	1.0160	1.0358	1.0586	–	0.9620	0.9792	1.0512	–
$6 \times 3 \times 1$	1.0231	1.0709	1.1856	–	0.9904	1.0589	1.1264	–
$6 \times 4 \times 1$	1.0231	1.0666	1.1431	–	1.0174	1.1024	1.2372	–
$6 \times 6 \times 1$	1.0701	1.3510	1.7346	2.2838	1.0444	1.1539	1.3007	1.7721
$8 \times 8 \times 1$	1.1220	1.7289	3.4191	5.0691	1.1446	1.4350	3.5168	4.7495

The selected configurations are $(3 \times 4 \times 1, R=1, h=1.80, \bar{h}=0.25)$ for CFP and $(6 \times 2 \times 1, R=1, h=0.85, \bar{h}=0.30)$ for ROI, with standardized-scale RMSEs of 0.9890 and 0.9620, respectively. Figure 3 shows the selected grids for a representative paired visit.

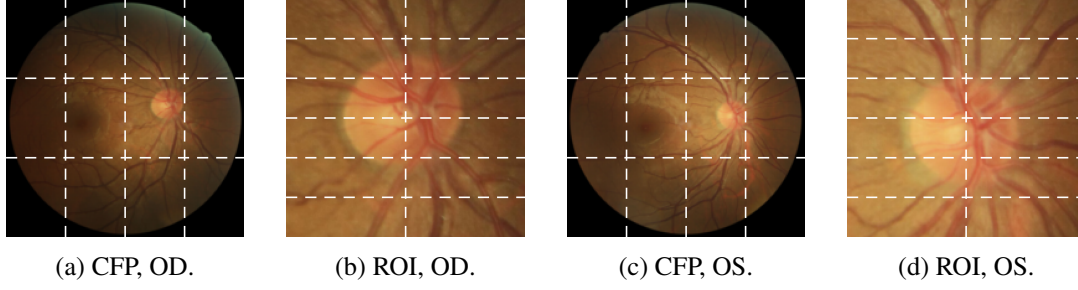


Fig 3: Representative paired-eye GRAPE images with the selected partition.

We next interpret the fitted ROI model. Figure 4 compares the independence and paired estimates of the 12 entries of the reduced coefficient matrix $\mathbf{A}(t)$, namely $a_{1s}(t)$, $s = 1, \dots, 12$. The pronounced age variation in the paired estimates supports the hypothesized age dependence of the association between ROI image features and IOP, while the differences between the two sets of curves show that accounting for within-pair dependence can materially alter the estimated image effects.

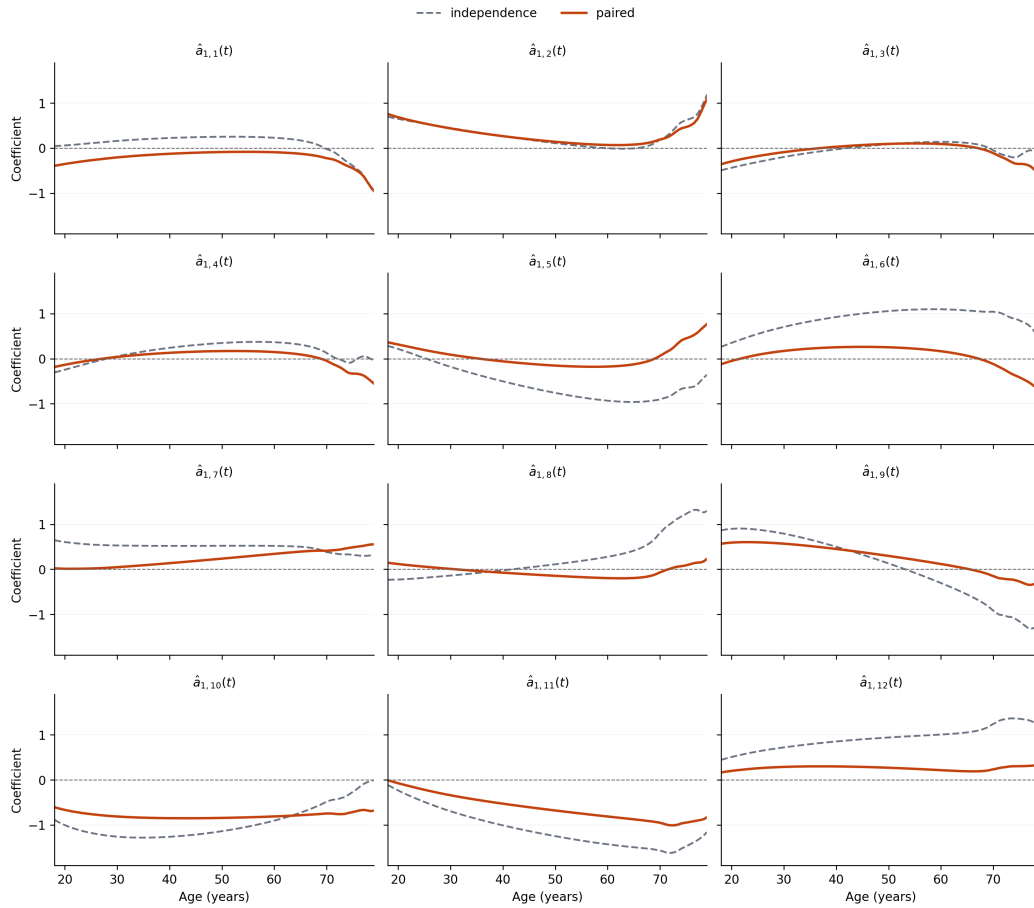


Fig 4: Independence and paired estimates of the entries $a_{1s}(t)$, $s = 1, \dots, 12$, of the reduced coefficient matrix $\mathbf{A}(t)$ for the ROI model.

Figure 5 shows the estimated residual standard-deviation function and the VF locations highlighted by the scalar-covariate analysis. The estimated residual variability rises gradually before age 30, increases more markedly between approximately ages 30 and 45, and remains broadly stable through age 60, suggesting modest age-dependent heteroscedasticity. The dip near age 70 and the subsequent rebound should be interpreted cautiously because observations are sparse near the upper age boundary, where kernel estimates are more sensitive to individual subjects. The fitted within-pair correlation is $\hat{\rho} = 0.617$, providing direct empirical support for the proposed paired covariance adjustment.

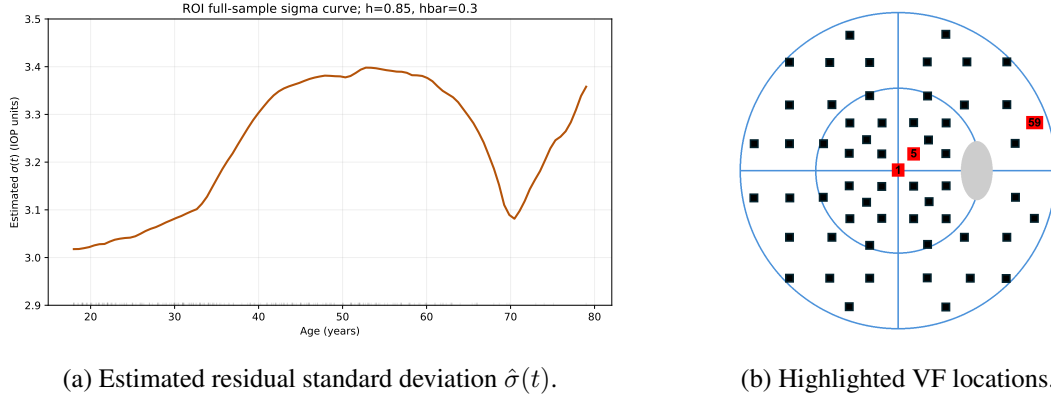


Fig 5: ROI residual standard deviation and highlighted VF locations.

Table 10 reports the complete scalar-coefficient results based on $B = 2000$ bootstrap replications. The gender effect is not significant, while the confidence intervals for VF00, VF04, and VF60 exclude zero. Together with the age-varying coefficient curves and the estimated positive within-eye-pair correlation, these results show that the proposed method identifies age-dependent image effects, clinically localized VF associations, and residual paired-eye dependence within a unified model.

6. Conclusion. Tensor regression techniques are increasingly recognized as powerful tools across a range of modern applications. Beyond the image data analysis presented in this study, these methods hold great potential in handling complex biomedical data structures, such as those arising in microbiome research or genetic network studies. In scenarios where intricate interaction effects are present, the varying coefficient tensor regression model proposed in this paper offers a flexible and effective framework. Its ability to accommodate high-dimensional and structured covariates makes it particularly well-suited for analyzing such complex data sources.

This study also has strong clinical relevance. The diagnostic insights derived from our analysis can assist ophthalmologists in making more accurate assessments of glaucoma status based on color fundus images, thereby enabling more precise and personalized treatment recommendations for patients. Currently, deep learning methods are widely adopted in ophthalmic image analysis for similar purposes. An exciting future direction would be to integrate our proposed model within a deep learning framework, which could enhance computational efficiency while producing interpretable and clinically actionable outputs.

Table 10: Scalar-coefficient estimates for the ROI paired-eye VCTR model based on $B = 2000$ bootstrap replications.

Covariate	Estimate	Bootstrap SE	95% CI	p-value
Gender	0.3636	0.7484	[-1.0519, 1.8401]	0.543
VF00	-4.7368	1.8050	[-7.7807, -0.7137]	0.023
VF01	2.4040	1.6152	[-1.0620, 5.2806]	0.148
VF02	1.2436	2.0067	[-2.9930, 4.9205]	0.522
VF03	3.8123	2.2362	[-0.3617, 8.3195]	0.070
VF04	4.6147	2.0020	[0.7861, 8.5312]	0.019
VF05	-3.2855	2.2925	[-8.2329, 1.0029]	0.112
VF06	-1.2673	2.3895	[-6.5481, 2.9152]	0.447
VF07	1.9111	2.8446	[-3.6889, 7.6151]	0.570
VF08	-2.1027	2.1817	[-6.6136, 1.7527]	0.295
VF09	0.6285	2.1515	[-3.4967, 5.0914]	0.689
VF10	-0.1016	1.7733	[-3.7164, 3.2306]	0.906
VF11	0.9169	1.7497	[-2.4376, 4.6234]	0.502
VF12	-0.7978	2.2431	[-4.4937, 4.3516]	0.886
VF13	3.9601	2.2563	[-0.8856, 7.9618]	0.119
VF14	0.4009	2.0320	[-1.8102, 6.2806]	0.758
VF15	-1.7483	2.6266	[-7.2105, 3.0866]	0.454
VF16	4.7956	2.7778	[-0.0749, 10.6414]	0.053
VF17	-3.2574	2.9900	[-9.2973, 2.1048]	0.286
VF18	-2.7720	2.3446	[-7.2509, 2.0119]	0.253
VF19	2.8077	2.2909	[-1.8514, 6.9746]	0.252
VF20	-0.5433	2.3672	[-5.0245, 4.2213]	0.826
VF22	1.8373	1.8401	[-1.7735, 5.5462]	0.298
VF23	-0.7241	1.5016	[-3.6695, 2.1683]	0.624
VF24	-2.4735	2.0545	[-6.5204, 1.3948]	0.229
VF25	-0.0420	2.6614	[-5.2189, 5.5348]	0.986
VF26	-1.9174	2.0246	[-5.9303, 2.3030]	0.345
VF27	-4.6170	2.7248	[-9.7895, 1.1205]	0.096
VF28	4.5282	2.9070	[-2.3001, 8.8359]	0.202
VF29	-4.6475	2.6019	[-9.7920, 0.3106]	0.067
Covariate	Estimate	Bootstrap SE	95% CI	p-value
VF30	-1.5992	1.8574	[-5.5043, 1.6875]	0.294
VF31	0.7623	2.0298	[-2.7546, 5.1378]	0.606
VF33	-1.4154	1.2216	[-3.9614, 0.6820]	0.190
VF34	0.9300	1.1479	[-1.4252, 3.1472]	0.466
VF35	-2.4346	1.5201	[-5.1943, 0.7145]	0.144
VF36	0.7484	1.4691	[-2.2744, 3.6347]	0.676
VF37	-1.1999	2.1288	[-5.5216, 2.9894]	0.556
VF38	1.5452	1.8560	[-2.1044, 5.1512]	0.386
VF39	-2.2188	1.7798	[-5.7668, 1.3339]	0.224
VF40	0.8376	1.5046	[-1.9421, 3.9402]	0.549
VF41	-0.9182	1.6364	[-4.0978, 2.4663]	0.556
VF42	0.6432	2.4218	[-3.5035, 6.0623]	0.706
VF43	0.7816	2.1333	[-2.8436, 5.5297]	0.600
VF44	-2.1332	1.9133	[-5.2422, 2.0390]	0.341
VF45	1.2080	1.5056	[-1.4518, 4.4074]	0.352
VF46	-2.2190	1.8511	[-5.7743, 1.3990]	0.222
VF47	2.2628	2.2334	[-1.8410, 7.0308]	0.241
VF48	1.2110	2.1274	[-2.6759, 5.6408]	0.504
VF49	-1.9245	1.7834	[-5.6493, 1.4426]	0.227
VF50	-0.2721	1.2640	[-2.5242, 2.3640]	0.928
VF51	-0.4740	1.0120	[-2.6796, 1.2211]	0.508
VF52	-1.3907	1.1003	[-3.7335, 0.5389]	0.167
VF53	0.5845	1.5011	[-2.2244, 3.7814]	0.629
VF54	1.6629	1.9435	[-2.1976, 5.2682]	0.457
VF55	0.8503	2.2200	[-3.6169, 5.1068]	0.795
VF56	0.4785	2.1697	[-4.4390, 4.2848]	0.926
VF57	1.9185	1.8760	[-2.3761, 4.8878]	0.422
VF58	0.8606	1.8375	[-2.9659, 4.2252]	0.752
VF59	1.1116	1.8124	[-2.2541, 4.8736]	0.488
VF60	2.4809	1.3808	[0.0853, 5.3715]	0.042

SUPPLEMENTARY MATERIAL

Supplementary Material for "Identification of Age-dependent Effects of Fundus Images on Intraocular Pressure for Glaucoma Patients: A Tensor Regression Approach". Section A contains the additional simulation results. Section B contains the additional real data experiment results. In Section C, we present the statistical theory, showing the asymptotic distribution and consistency for estimated functions and coefficients. And Section D provides formal proofs for all theoretical results.

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